Constructor Theory and Software in the Natural World

Eine konstruktivistische Theorie zur Entstehung von präbiotischer Information

Was ist Information?

Braucht Informationsentstehung die Biologie oder den Menschen?

Oder entstand Information vielleicht zuerst?

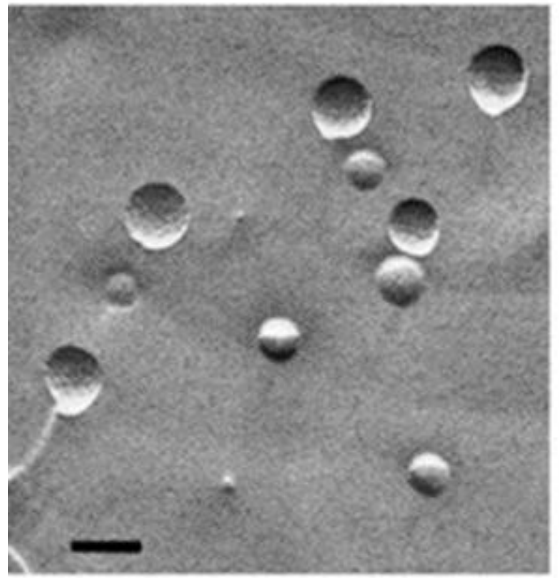
**Und was hat das alles mit Geckos,
präbiotischen Protozellen und Quantenfluktuationen zu tun?**

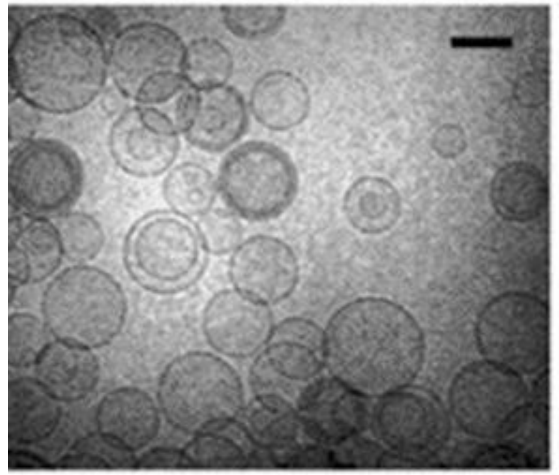


Michael Massoth
Hochschule Darmstadt (h_da)
Master-Hauptseminar 14.10.2026

Erste Protozellen entstanden vor 4,2 Milliarden Jahren



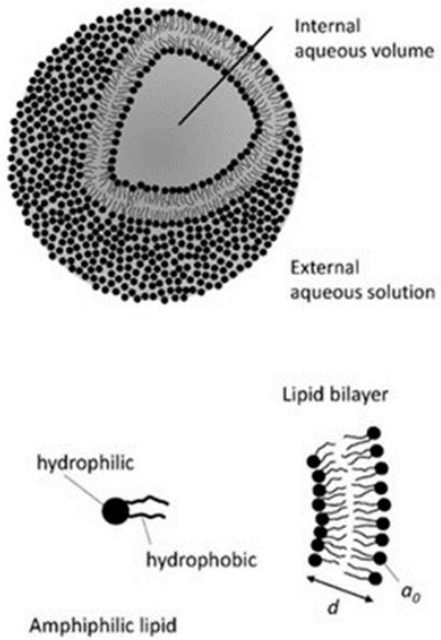


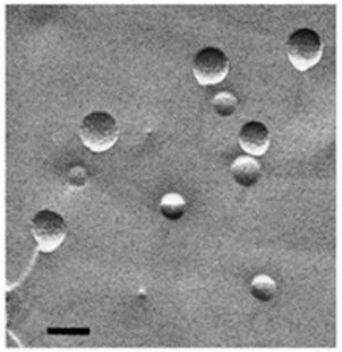


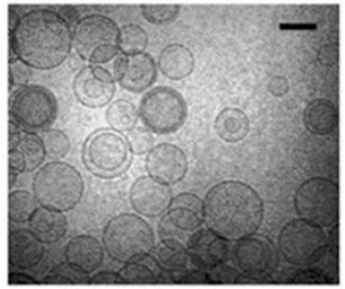
Meine Forschungsfrage(n):

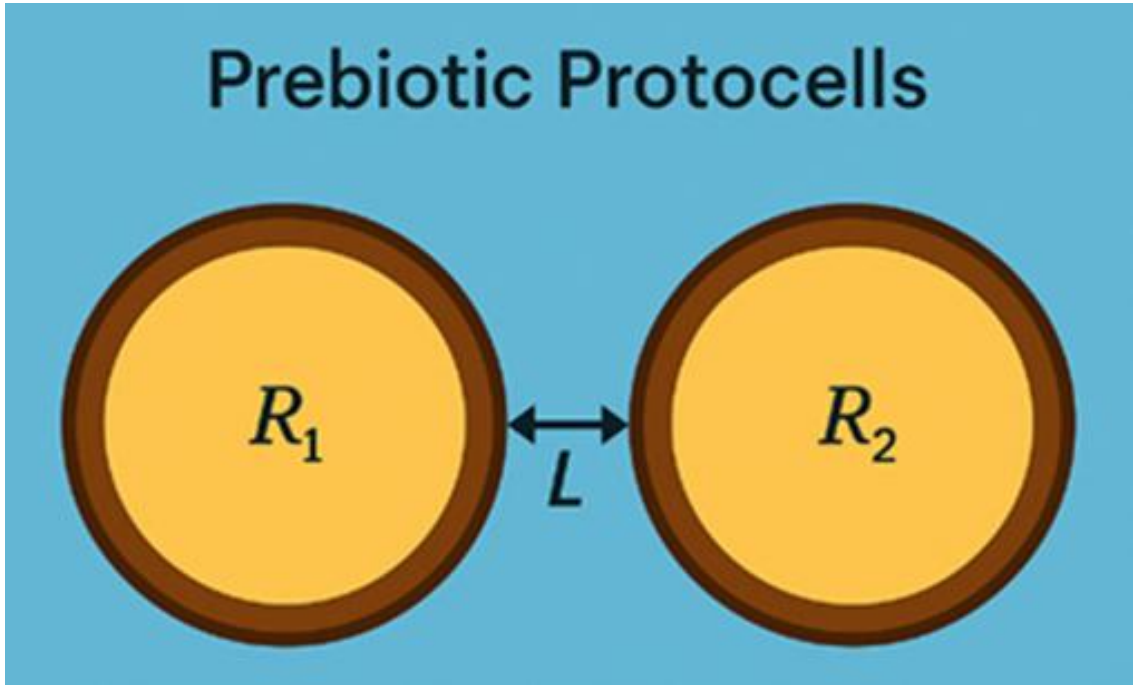
Wie entstand Information in einer präbiotischen Welt ohne Gene?

Konnten bereits Physik und Informatik alleine (!), Information, Funktionalität und Bedeutung in einer Welt ohne Biologie hervorbringen?

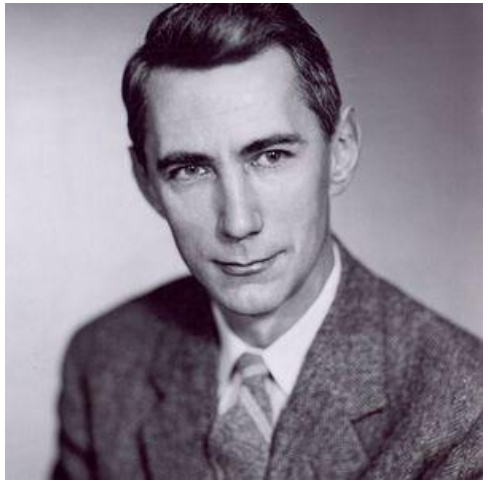








Was ist Information?



Information ist das Maß für die **Reduktion von Unsicherheit** über den Zustand eines Systems oder die empfangene Nachricht.

Betrachtet wird dabei nur die Syntax (Zeichen, Symbole, Wahrscheinlichkeiten, Shannon-Entropie, …)

(Shannon, 1948)





David Deutsch & Chiara Marletto (Constructor Theory of Information, 2015):

Information existiert dort, wo die Naturgesetze **wiederholbare Aufgaben** zum **Unterscheiden**, **Transformieren** und **Kopieren** physikalischer Zustände als möglich zulassen. [Task, Constructor, Substrat]

Construction Theory of Information

Ein physikalischer Ansatz zur Bestimmung, welche Informationsaufgaben möglich sind

1. Task (Aufgabe)

Eine Transformation $x \to y$ eines physikalischen Systems.



2. Possible / Impossible

Die zentrale Frage: Welche Tasks sind nach den Naturgesetzen möglich, welche unmöglich?



3. Constructor

Ein System, das eine Aufgabe wiederholt ausführen kann, ohne seine Fähigkeit dazu zu verlieren.



4. Substrate

Das physikalische System, auf dem eine Aufgabe ausgeführt wird.



5. Attribute

Eine physikalische Eigenschaft oder ein Zustand eines Substrats.



6. Variable

Eine Menge möglicher Attribute eines Systems.

$X = \{x_1, x_2, \ldots, x_n\}$



7. Information Medium

Ein physikalisches System, auf dem Informationsaufgaben möglich sind.

8. Information Variable

Eine Menge von Zuständen, die unterschieden, transformiert und kopiert werden können.

$V = \{v_1, v_2, \ldots, v_n\}$



9. Distinguishability

Zustände können zuverlässig unterschieden werden.



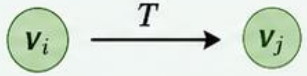
10. Copyability

Zustände können auf andere Substrate übertragen bzw. kopiert werden.



11. Computation

Zustände können gemäß erlaubter Transformationen ineinander überführt werden.



12. Interoperability Principle

Information muss zwischen unterschiedlichen physikalischen Trägern übertragbar sein.



13. Measurement

Eine spezielle Task zur Unterscheidung von Zuständen.



14. Superinformation

Informationsträger, bei denen bestimmte Informationsaufgaben unmöglich sind (z. B. Quantensysteme).



- × perfektes Kopieren
- × perfekte Unterscheidung
- × bestimmte Transformationen

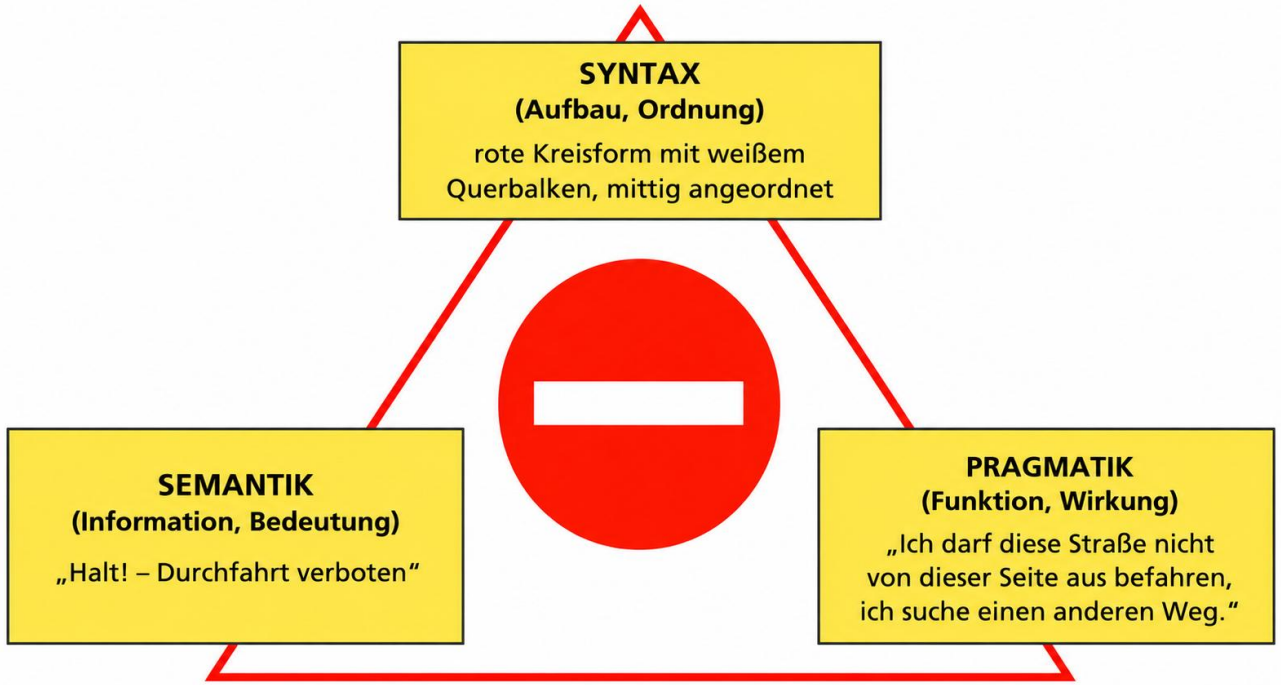
Kernidee

Information ist nicht an ein bestimmtes physikalisches System gebunden, sondern an die Möglichkeiten, zu unterscheiden, zu kopieren und zu transformieren – gemäß den Naturgesetzen.

Die Construction Theory of Information beantwortet die Frage:

Welche Informationsaufgaben sind möglich – und warum?

Syntax, Semantik und Pragmatik



Syntax beantwortet die Frage:
„**Welche Muster und Zustände existieren und wie hängen sie strukturell zusammen?**“

Semantik: „**Was bewirkt ein Zustand?**“

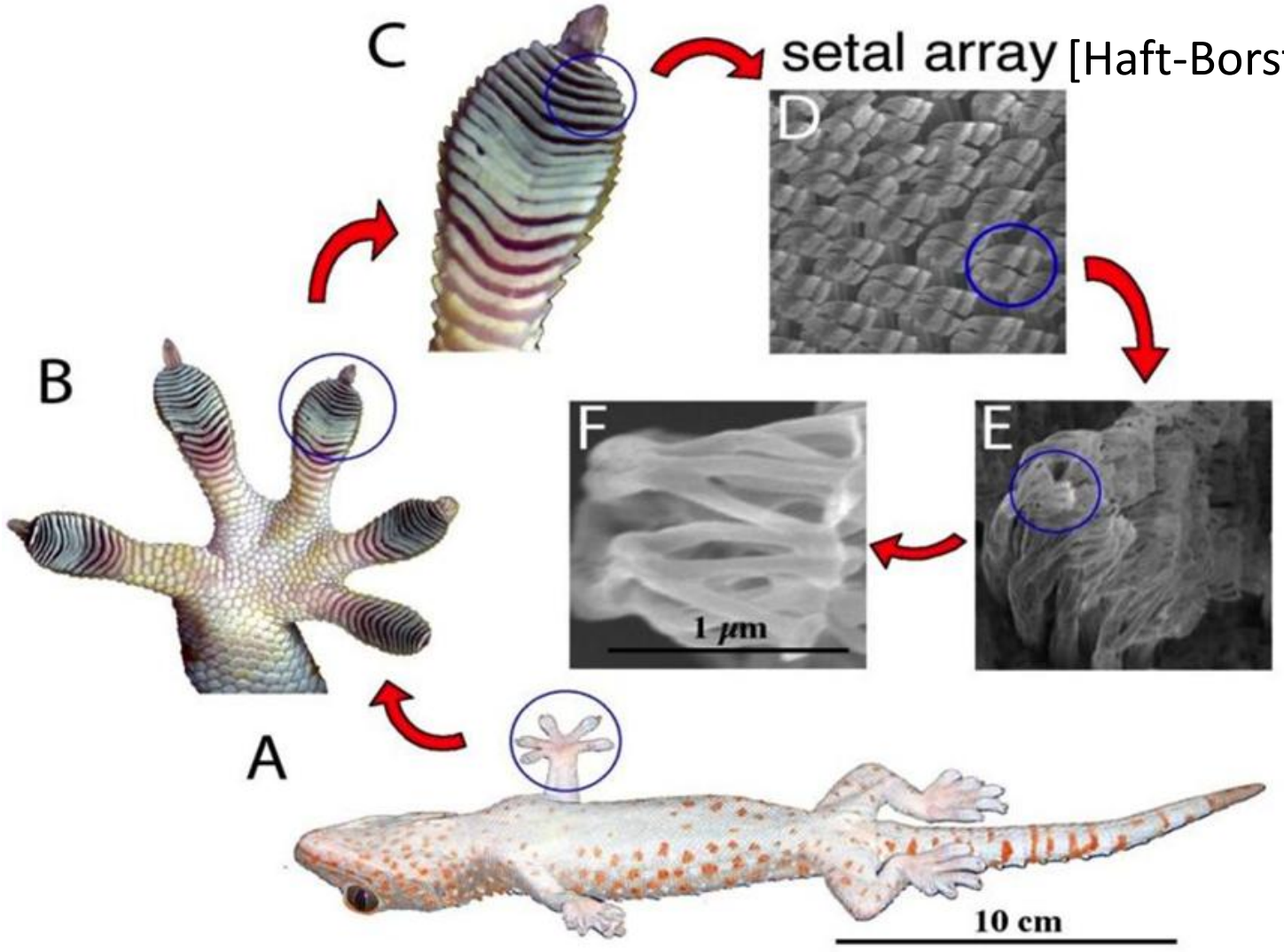
Pragmatik:
„**Wozu wird Information verwendet?**“

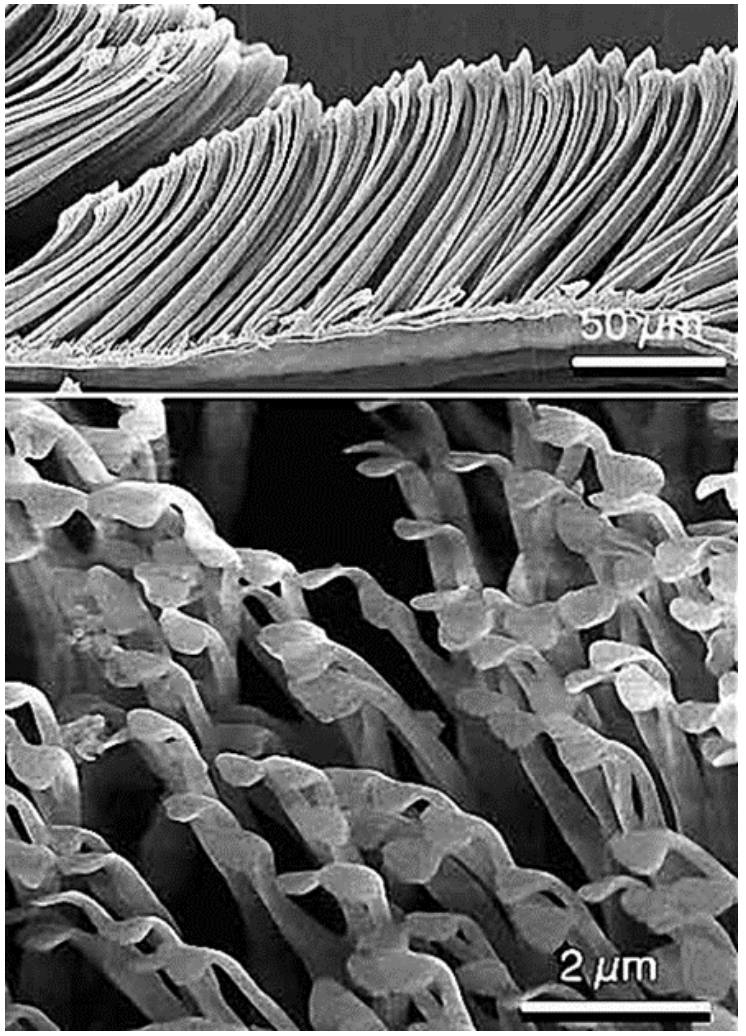
Syntax beschreibt Struktur und Regelmäßigkeit von Informationsmustern,
Semantik beschreibt Funktion/Bedeutung und
Pragmatik beschreibt die Nutzung der Funktion/Bedeutung zur Erzeugung von Wirkung.



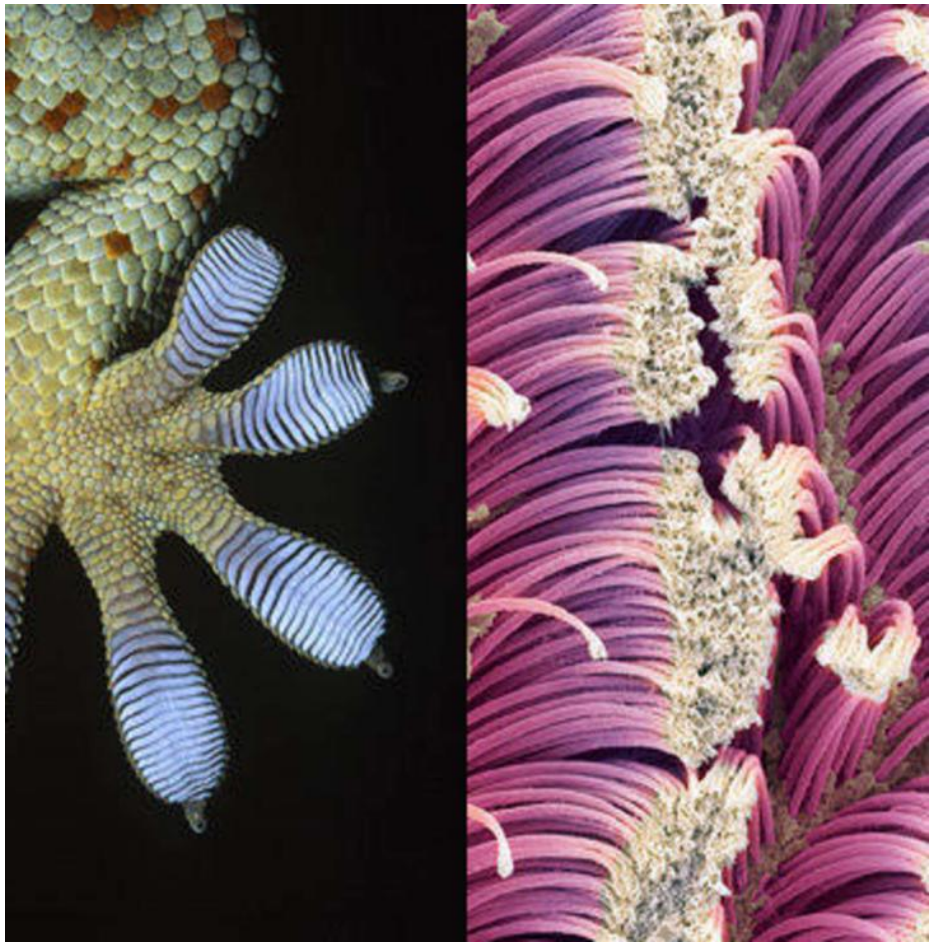
Wie schafft es ein Gecko senkrecht eine Wand hochzulaufen?

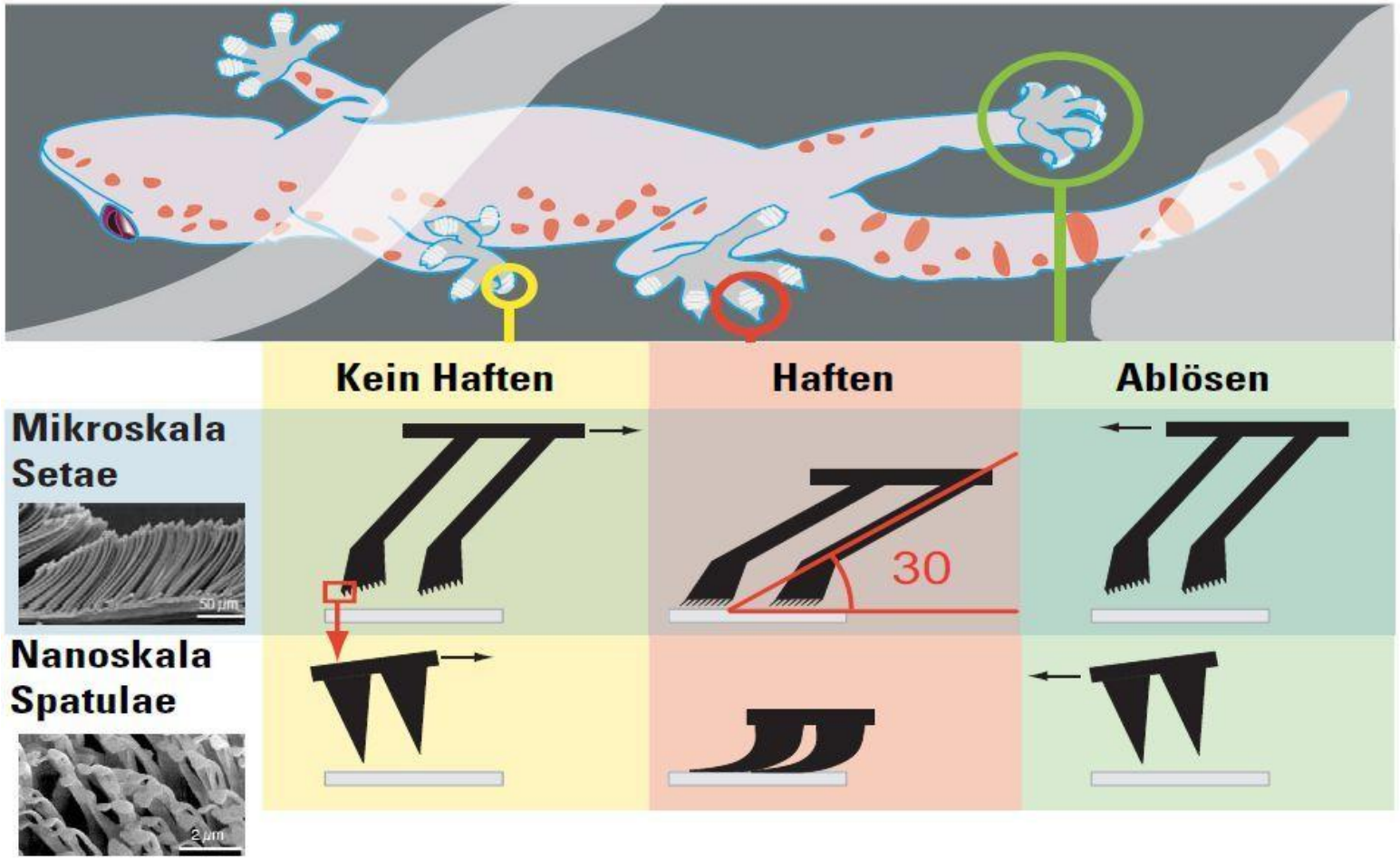






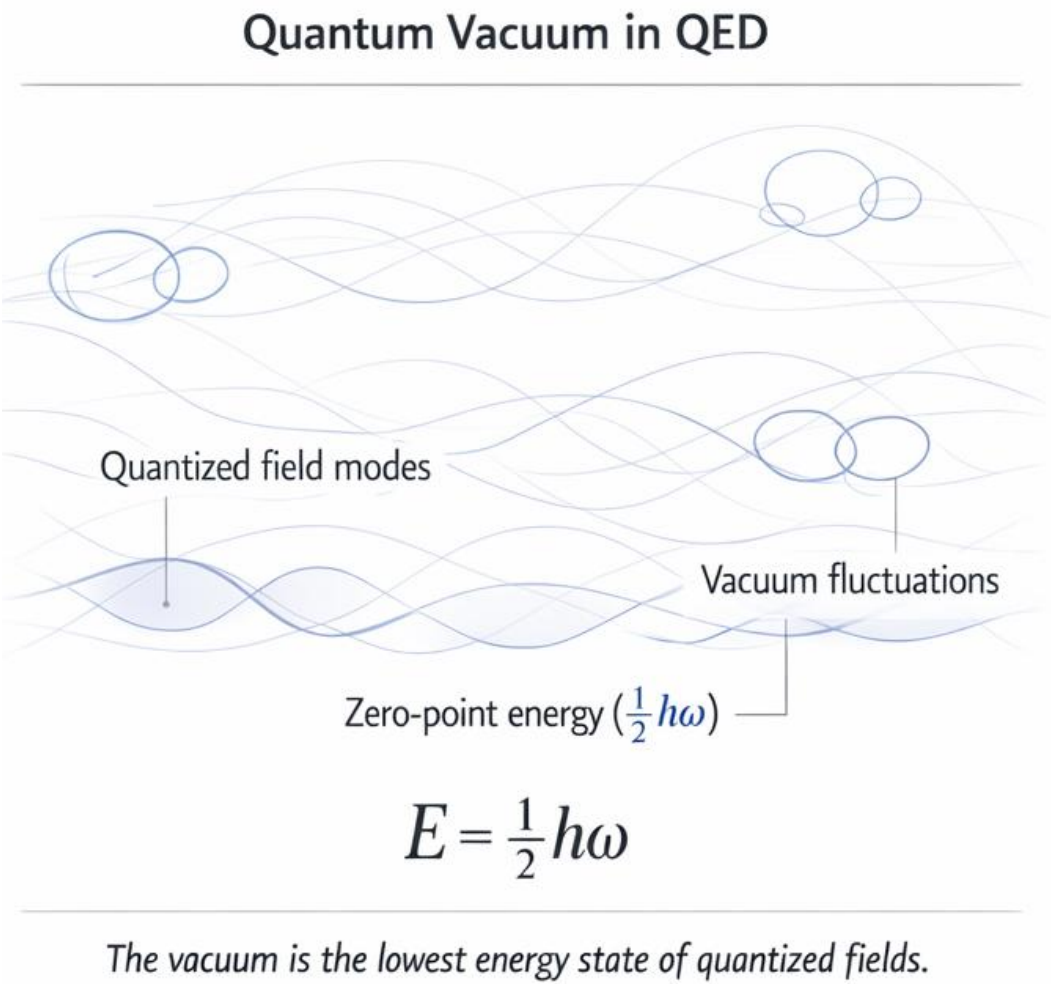
Der Gecko nutzt keine Klebstoffe oder Saugnäpfe. Er nutzt Quantenphysik.





Der Gecko nutzt keine Klebstoffe oder Saugnäpfe. Er nutzt Quantenphysik.

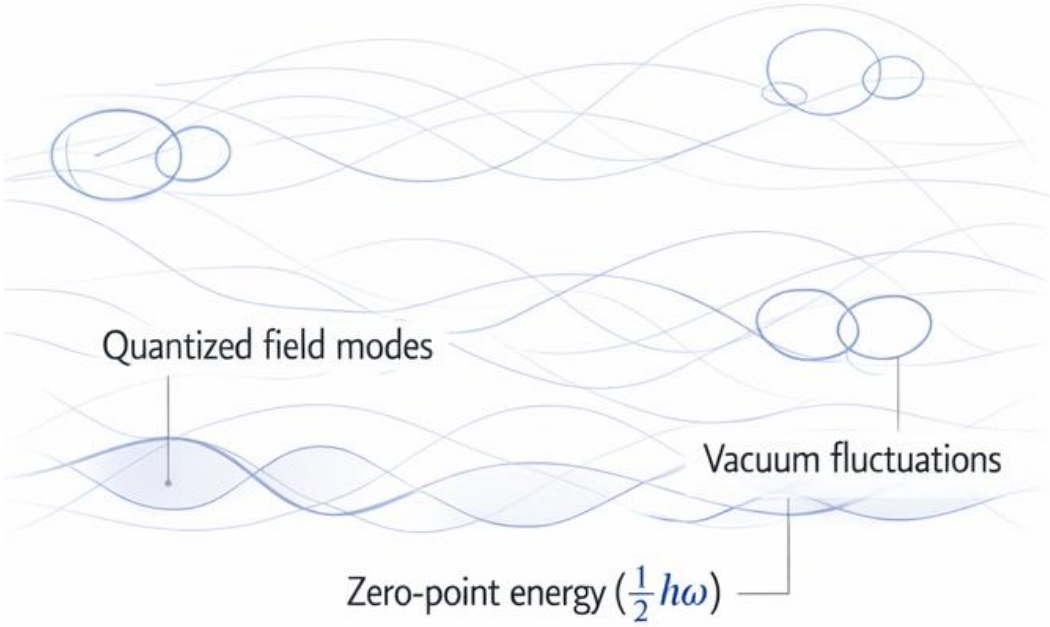
Some Physical Basics: The Quantum Vacuum in QED (1)



- Free of particles.
- The vacuum is not “nothingness,” but rather the state of **lowest energy state** of quantised fields.
- According to **Heisenberg's uncertainty principle**, a quantum oscillator can never come to a complete rest. It always vibrates.
  → **Vacuum fluctuations**.

Some Physical Basics: The Vacuum & Casimir-Effect (2)

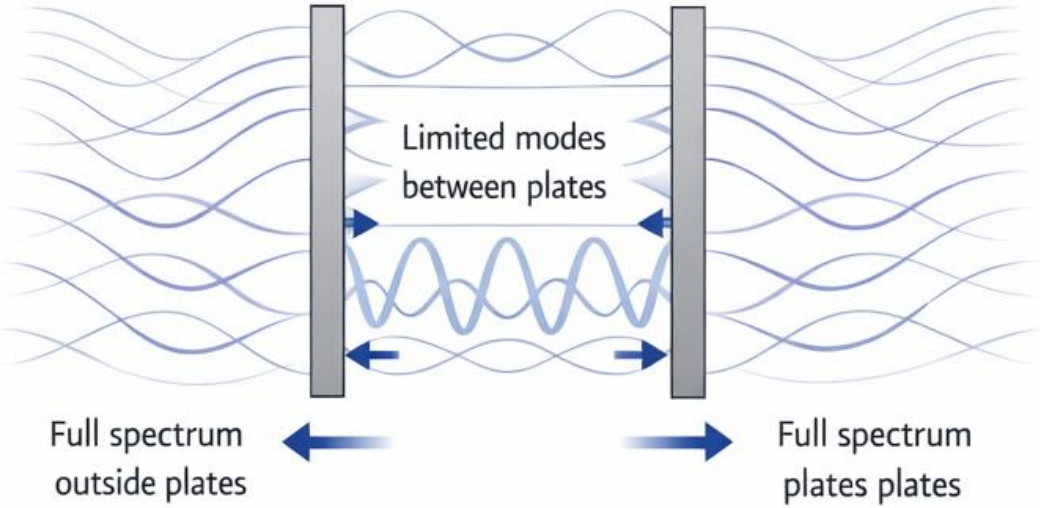
Quantum Vacuum in QED



$$E = \frac{1}{2}h\omega$$

The vacuum is the lowest energy state of quantized fields.

Casimir Effect



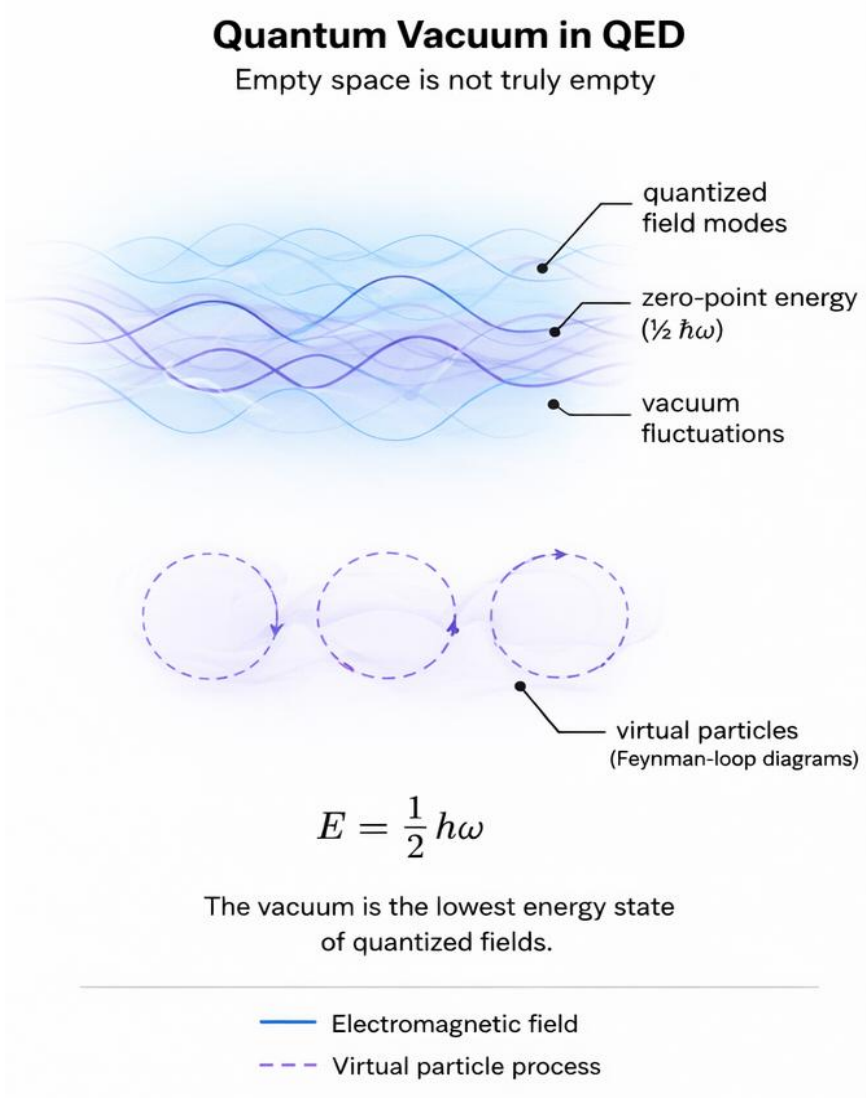
$$F = \frac{-\pi^2 hc}{240\,a^4}$$

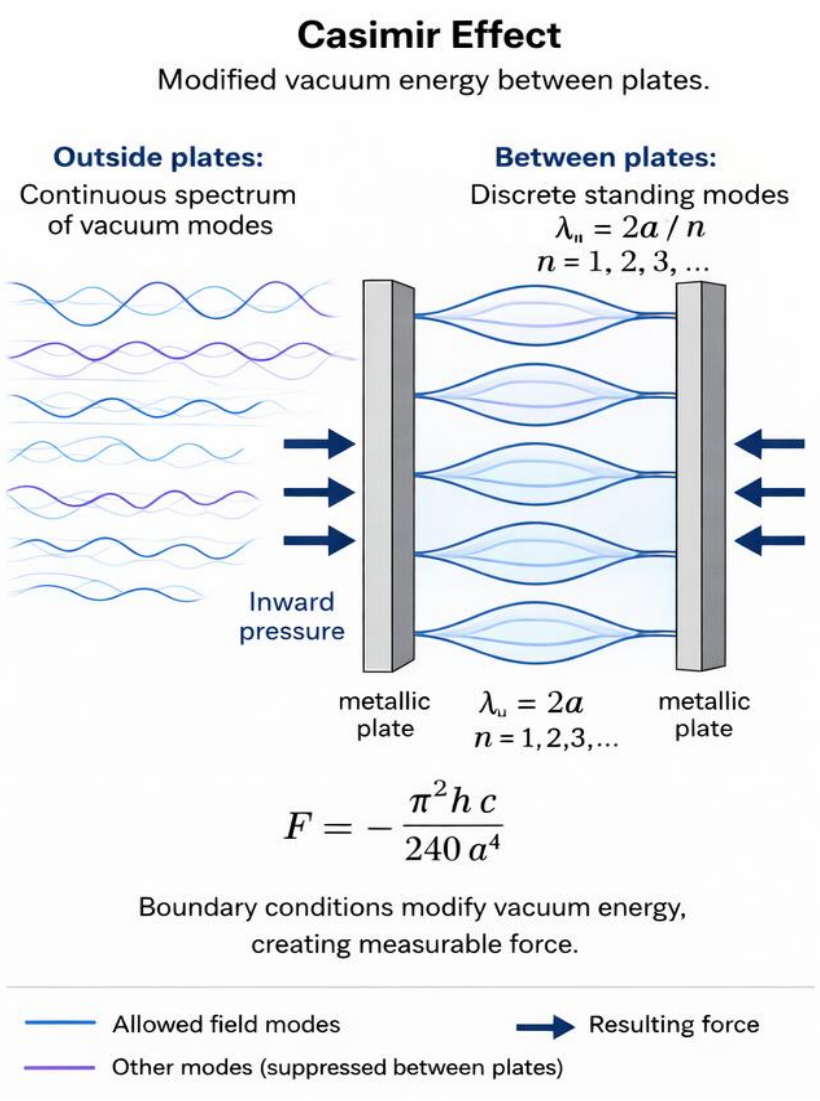
[Casimir, 1948]

Boundary conditions modify vacuum energy, creating measurable force.

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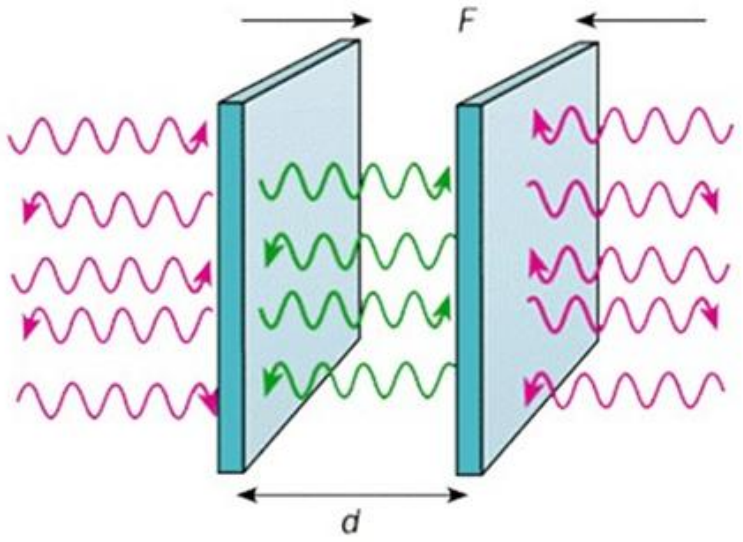
Some Physical Basics: The Vacuum & Casimir-Effect (3)



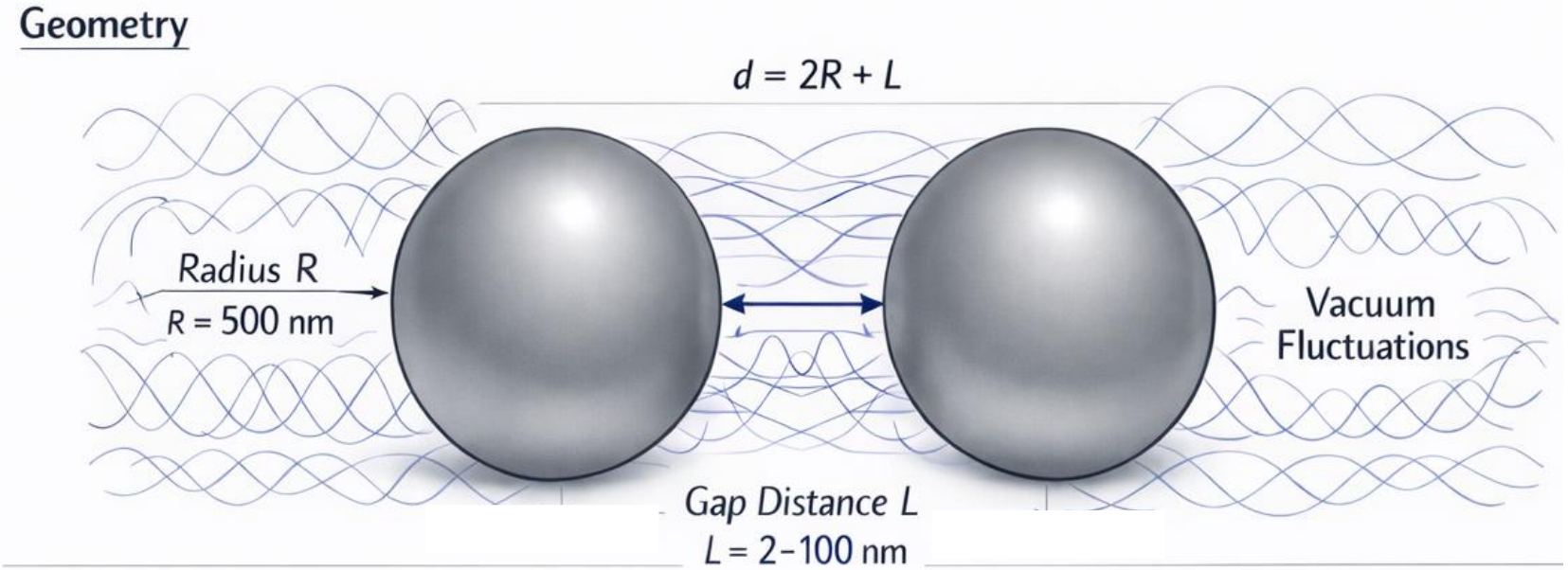


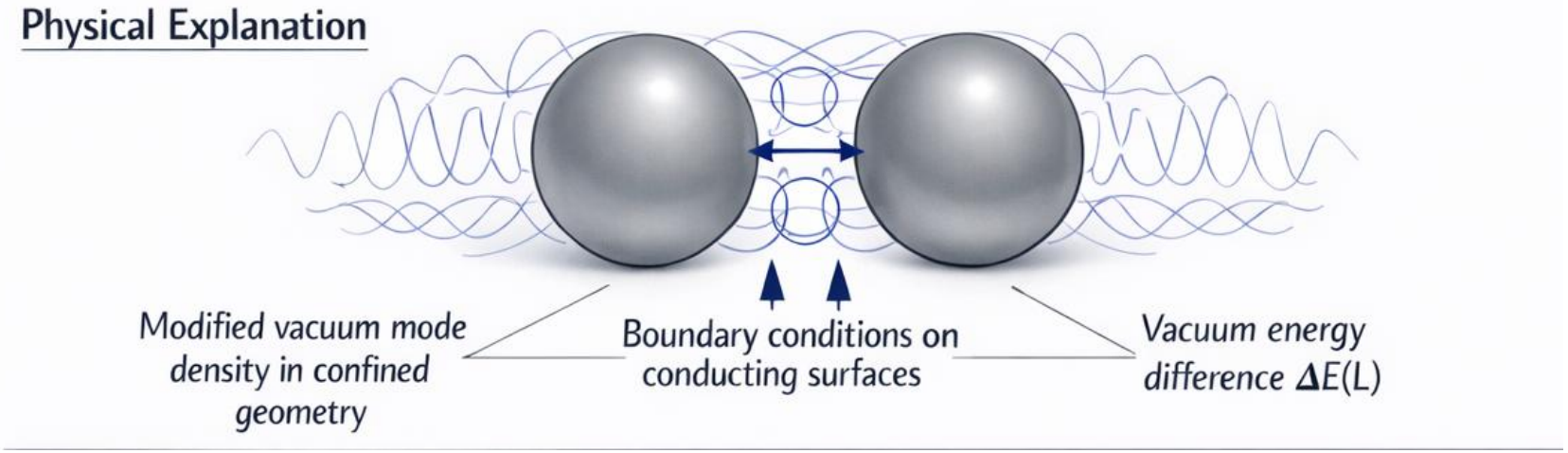
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Casimir-Effect between two metallic Spheres (4)

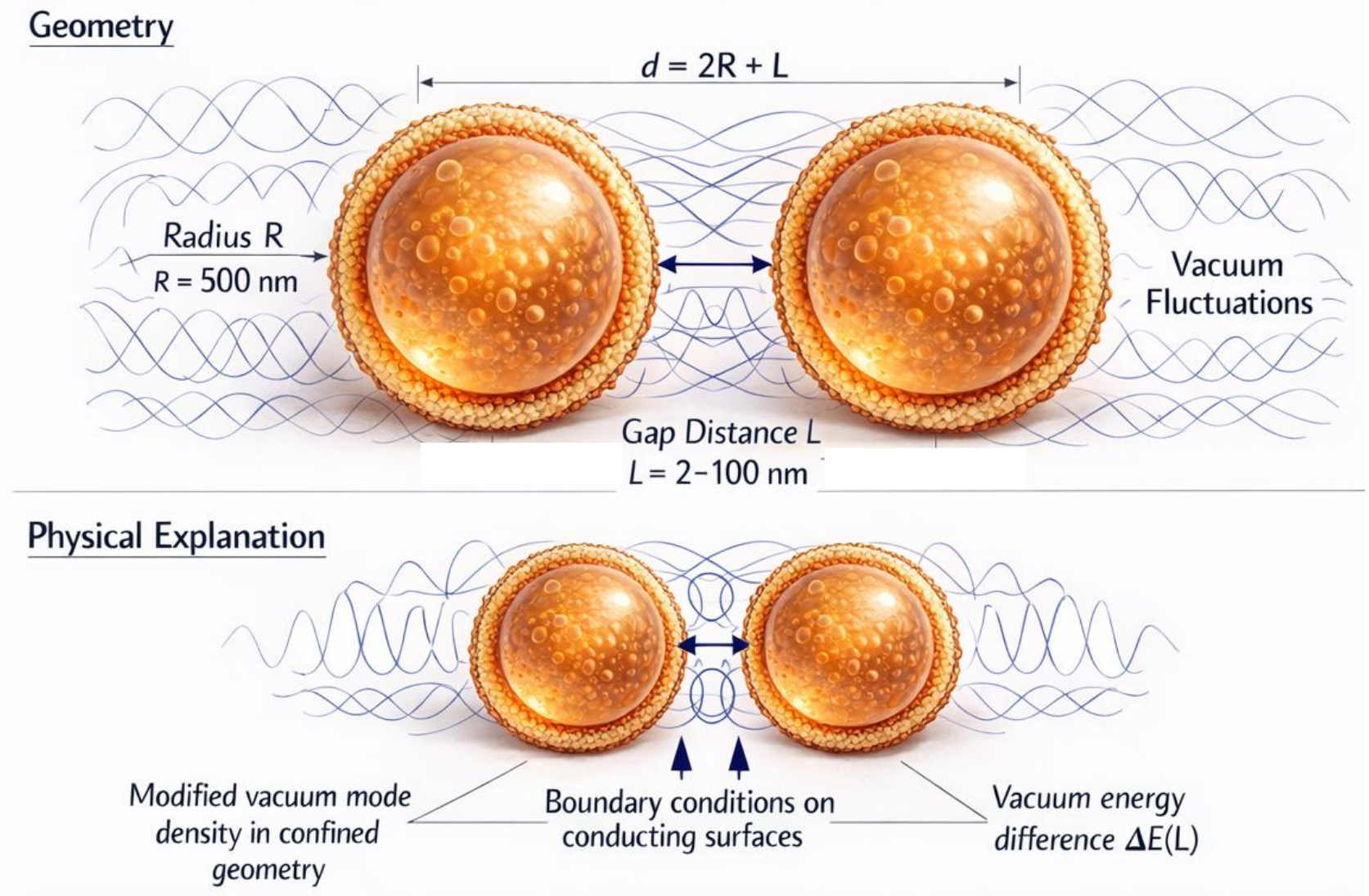


Experimenteller Nachweis:
Demonstration of the Casimir Force in the 0.6 to 6 µm Range, PRL,
Steve K. Lamoreaux, 1997



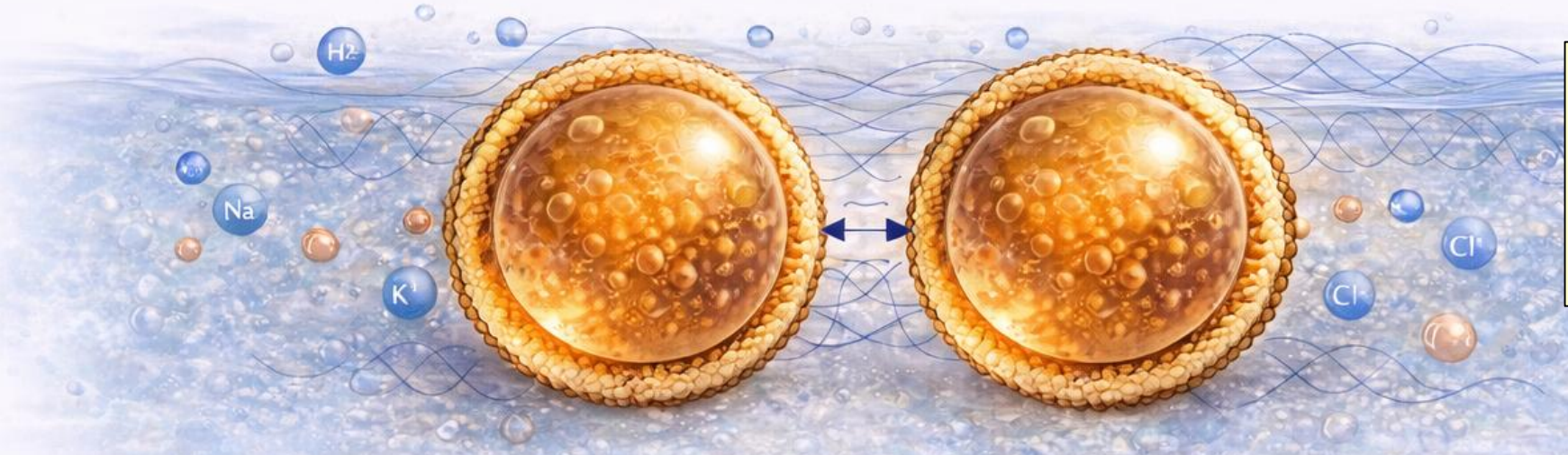


Casimir-Effect between two Protocells ??? (5)



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Casimir-Lifshitz-interactions in primordial soup (6)



Kationen-Durchmesser:
$Na^+ \approx 0{,}20$ nm
$Ka^+ \approx 0{,}28$ nm
$H_3O^+ \approx 0{,}56$ - $0{,}76$ nm

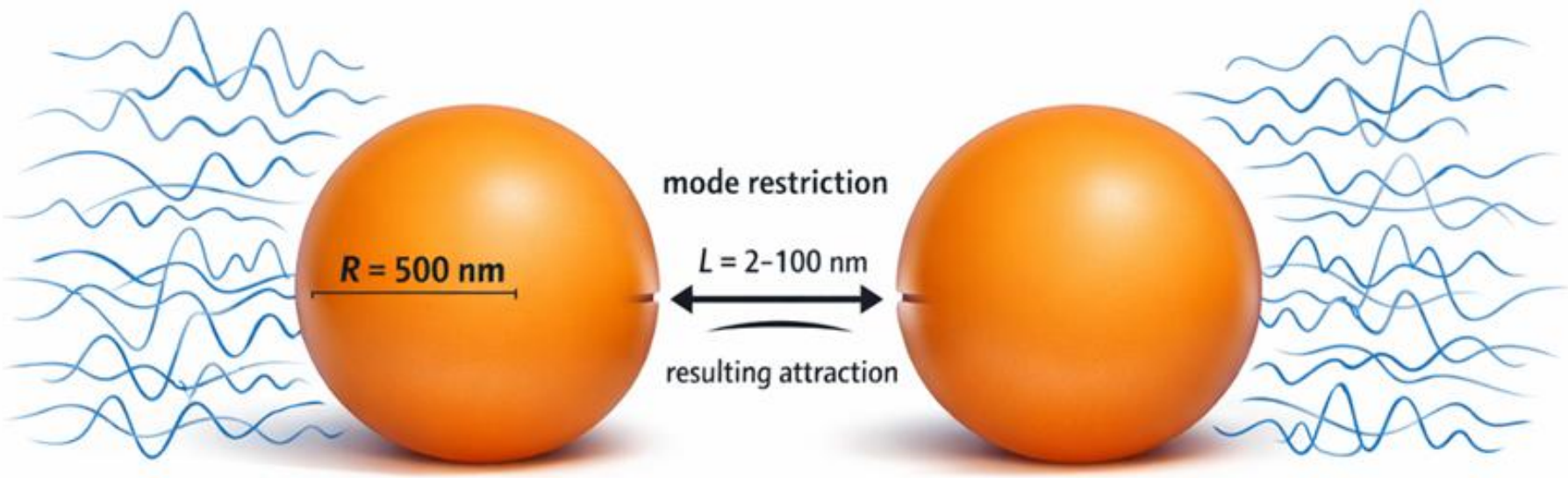
CONDITIONS

- Polarization (dielectric properties)
- Screening (Debye length λ_D)
- Higher ionic strength
  → smaller λ_D → stronger screening
- Modified Field Modes

Ideal Vacuum	Prebiotic Medium
Field Conditions:	(Water, ions)
Permittivity	Dielectric response (imaginary frequencies): $\epsilon_I(i\epsilon_{iII} = \epsilon_{II}(i\epsilon_i = \epsilon_4(\epsilon_i)$
Screening	Debye Length λ_D
Force Magnitude:	pikoNewton
Force is attractive	

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Casimir–Lifshitz Physics:



$$F_{CL}(L) \approx -\frac{A_{eff}}{6} \cdot \frac{R_{eff}}{L^2}$$

$$\text{with } R_{eff} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

A_{eff} = *effective Hamaker constant* $\approx 5 \times 10^{-21}$ Joule.

Ein dynamisches System mit Attraktoren

Universal Casimir interaction between two dielectric spheres in salted water

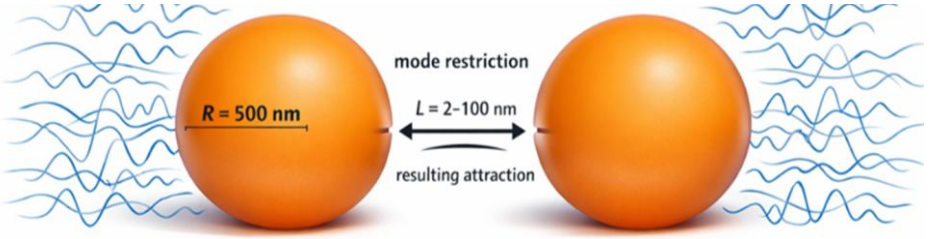
Tanja Schoger,[1] Benjamin Spreng,[2] Gert-Ludwig Ingold,[1,*] Paulo A. Maia Neto,[3,†] and Serge Reynaud[4,‡]

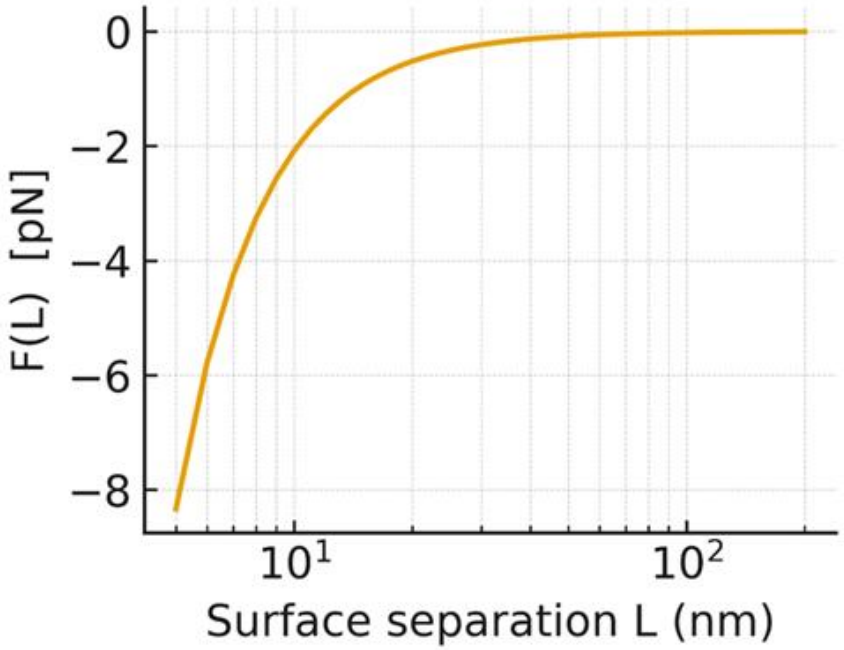
[1] *Universität Augsburg, Institut für Physik, 86135 Augsburg, Germany*
[2] *Department of Electrical and Computer Engineering, University of California, Davis, CA 95616, USA*
[3] *Instituto de Física, Universidade Federal do Rio de Janeiro Caixa Postal 68528, Rio de Janeiro, RJ, 21941-972, Brazil*
[4] *Laboratoire Kastler Brossel, Sorbonne Université, CNRS, ENS-PSL, Collège de France, Campus Jussieu, 75005 Paris, France*

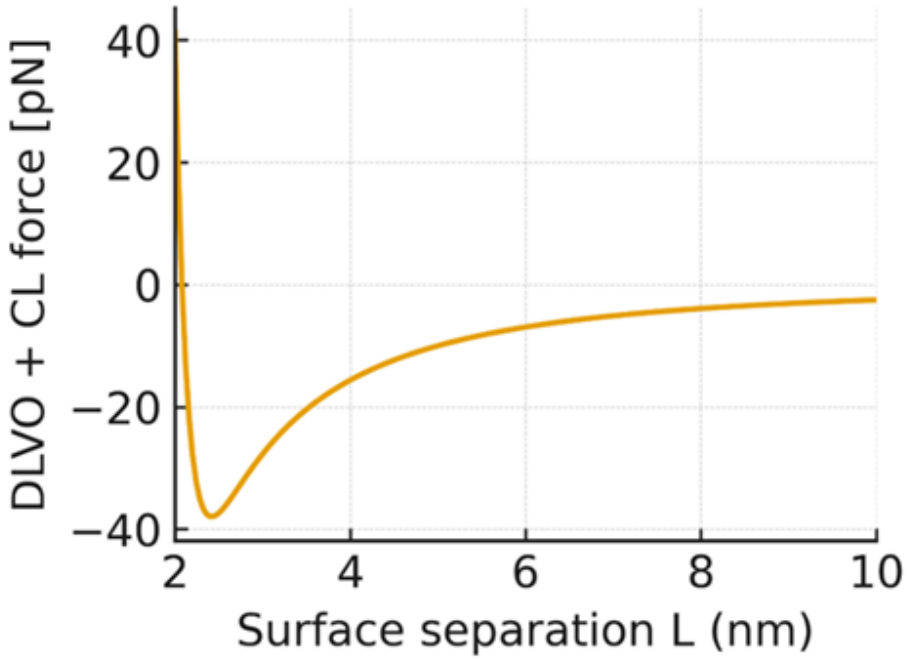
(Dated: December 17, 2021)

We study the Casimir interaction between two dielectric spheres immersed in a salted solution at distances larger than the Debye screening length. The long distance behavior is dominated by the non-screened interaction due to low-frequency transverse magnetic thermal fluctuations. It shows universality properties in its dependence on geometric dimensions and independence of dielectric functions of the particles, with these properties related to approximate conformal invariance.

Own Numerical Calculations:







Attractive Casimir–Lifshitz force $F_{CL}(L)$ *between two PMBC-like protocells* (R_1=R_2=500 nm) *as a function of separation L.*

Resulting total force *from* F_{DLVO} *and* F_{CL} *contributions over L=2–10 nm.*

Ein dynamisches System mit Attraktoren

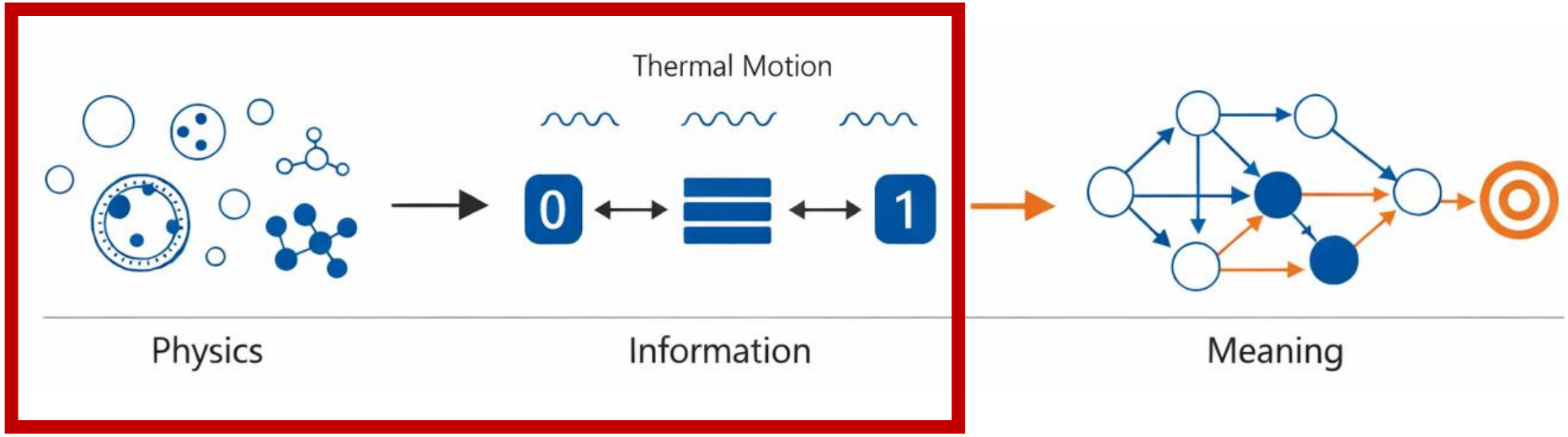
Unsere Cluster-Galerie: Diskrete unterscheidbare Makrozustände



N	Protocell Structure	N_{bonds}	$E_N/k_BT \approx$
2	Dimer	1	−5.1
3	Triangularer Trimer	3	−15.2
4	Tetrahedron	6	−30.5
6	Oktahedron	12	−61.0
7	Pentagonal Bipyramid	15	−76.2
13	Icosahedral 13-Cluster	42	−213.4

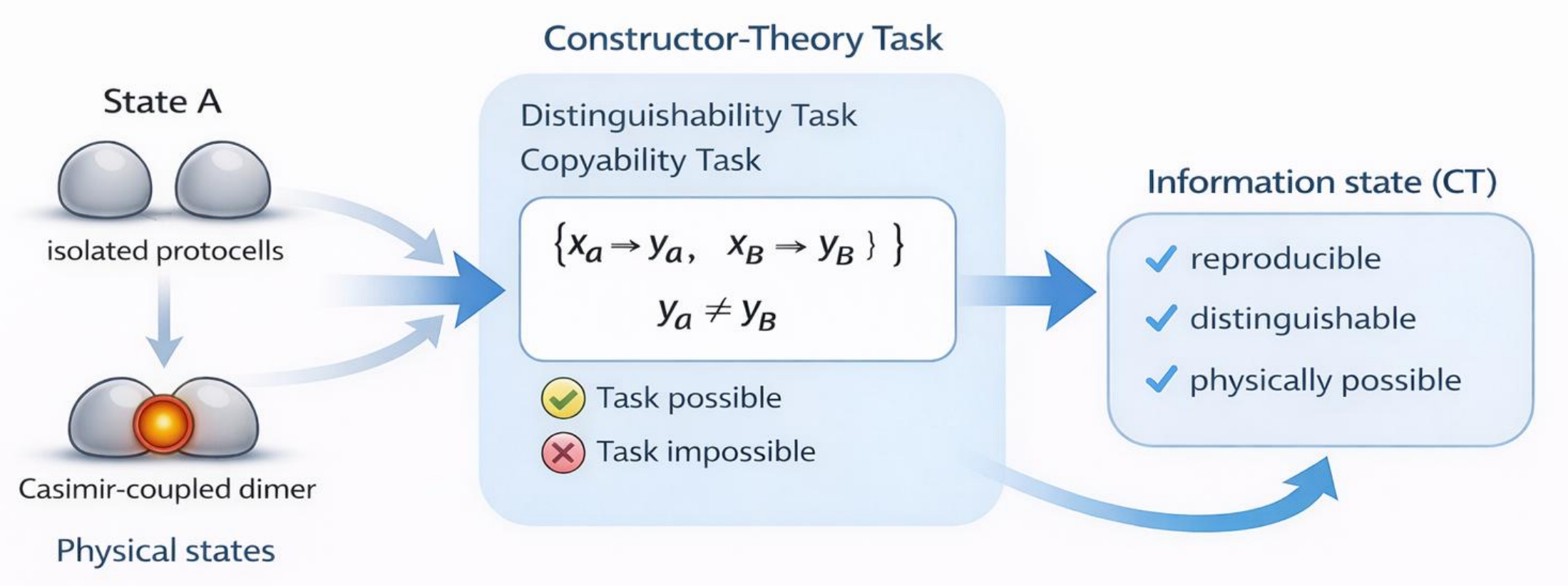
Das ist die **Syntax** der Protozellen: Die „Buchstaben“, „Symbole“ oder „Automaten“.

Von einem physikalischen Unterschied zu Information

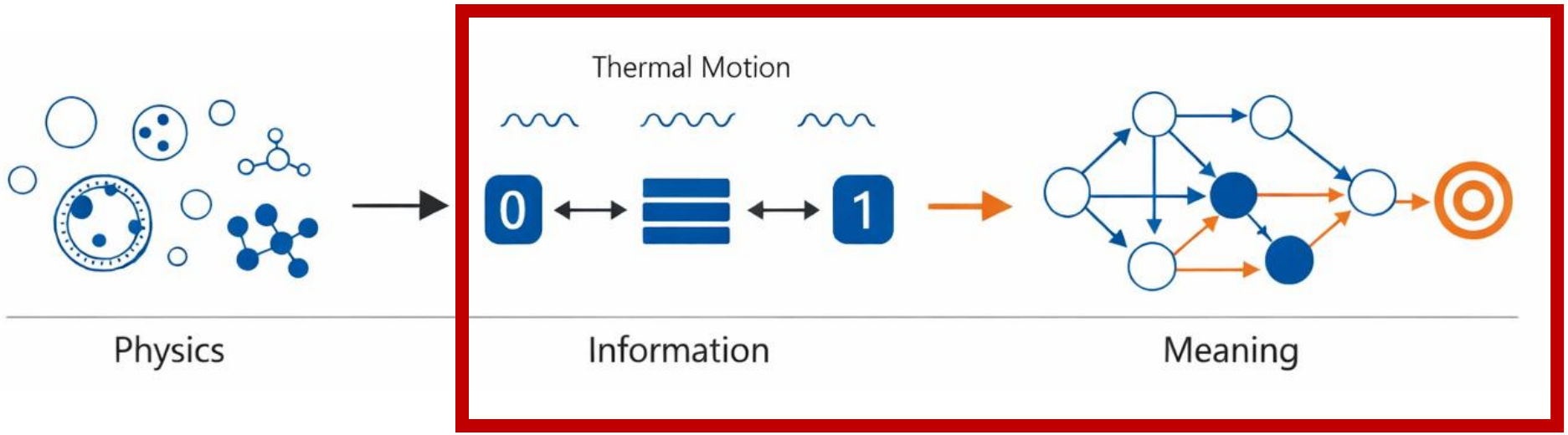




Von einem physikalischen Unterschied zu Information in der Constructor Theory



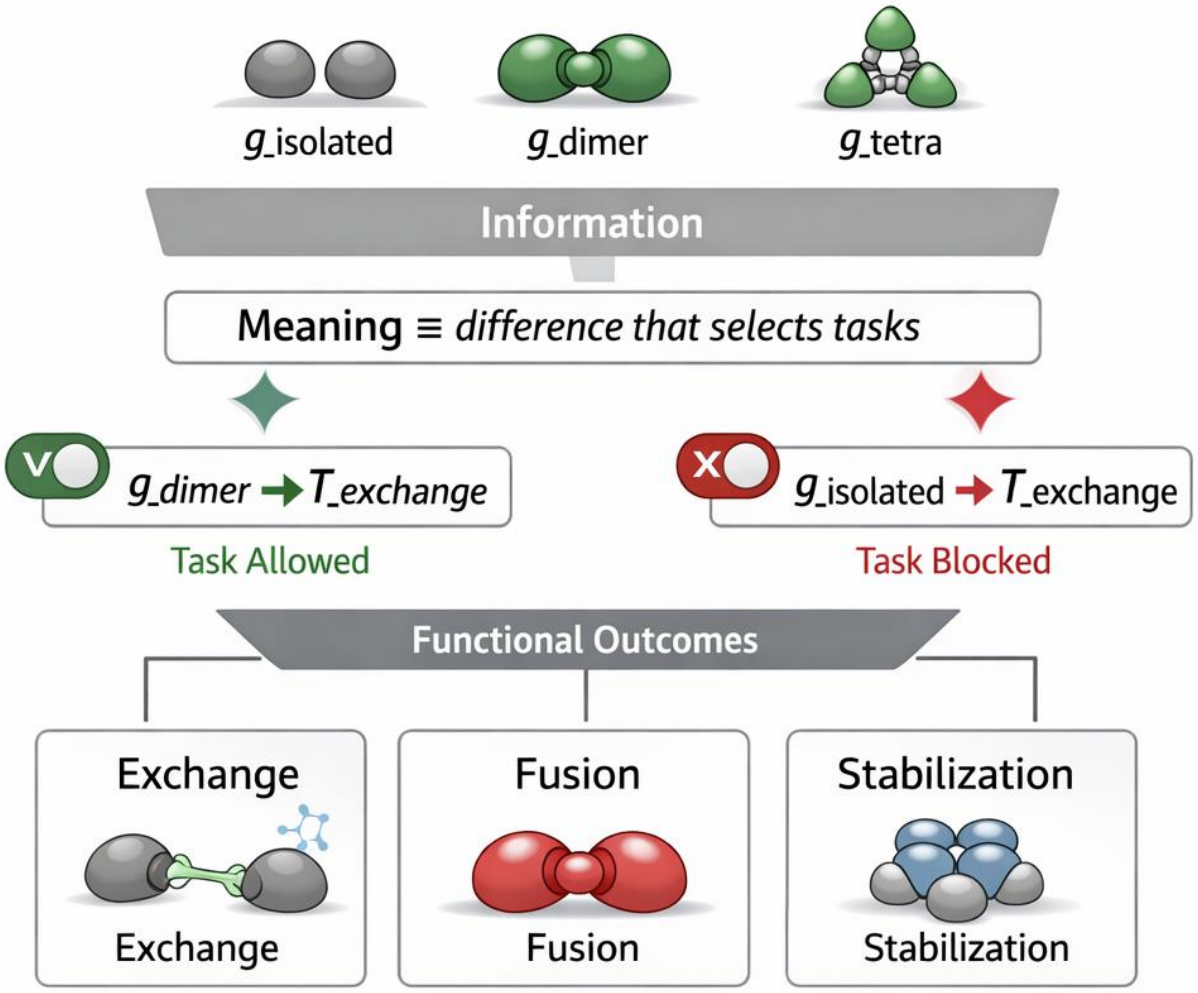
Von der Information zur Bedeutung





Von der Information zur Bedeutung durch Task-Gating in Protozell-Clustern

- Der Dimer-Zustand ermöglicht Austauschvorgänge (Ionen/Moleküle), die Monomere nicht bewältigen können.
- Trennungen mit geringem Abstand ermöglichen Fusionen/Hemifusionen leichter als Trennungen mit großem Abstand.
- Bestimmte Geometrien können Gradienten konzentrieren oder Reaktionsmikrozonen stabilisieren. Somit werden Geometrie und Abstand zu Kontrollvariablen – im Sinne der CT.



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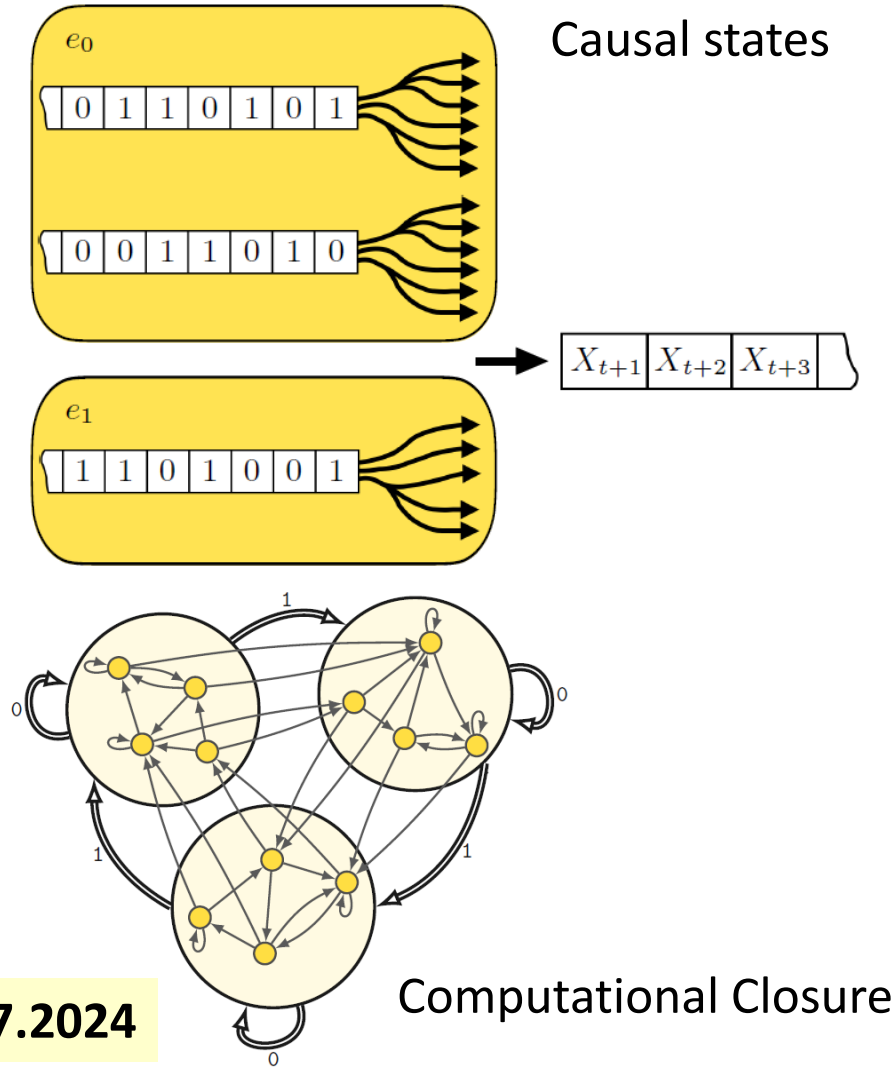
Rosas et al.: Software in the natural world (1)

Software in the natural world: A computational approach to hierarchical emergence

Fernando E. Rosas,[1,2,3,4,*] Bernhard C. Geiger,[5,6] Andrea I Luppi,[7]
Anil K. Seth,[1,2] Daniel Polani,[8] Michael Gastpar,[9] and Pedro A.M. Mediano[10,11]

[1] *Department of Informatics, University of Sussex*
[2] *Sussex Centre for Consciousness Science and Sussex AI, University of Sussex*
[3] *Center for Psychedelic Research and Centre for Complexity Science, Department of Brain Science, Imperial College London*
[4] *Center for Eudaimonia and Human Flourishing, University of Oxford*
[5] *Know-Center GmbH, Graz, Austria*
[6] *Signal Processing and Speech Communication Laboratory, Graz University of Technology, Graz, Austria*
[7] *Montreal Neurological Institute, McGill University*
[8] *Department of Computer Science, University of Hertfordshire, Hatfield, UK*
[9] *School of Computer and Communication Sciences, EPFL, Lausanne, Switzerland*
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[11] *Division of Psychology and Language Sciences, University College London*

Understanding the functional architecture of complex systems is crucial to illuminate their inner workings and enable effective methods for their prediction and control. Recent advances have introduced tools to characterise emergent macroscopic levels; however, while these approaches are successful in identifying *when* emergence takes place, they are limited in the extent they can determine *how* it does. Here we address this important limitation by developing a computational approach to emergence, which characterises macroscopic processes in terms of their computational capabilities. Concretely, we articulate a view on emergence based on how software works, which is rooted on a mathematical formalisation of how macroscopic processes can express self-contained informational, interventional, and computational properties. This framework reveals a hierarchy of nested self-contained processes that determines what computations take place at what level, which in turn delineates the functional architecture of a complex system. This approach is illustrated on paradigmatic models from the statistical physics and computational neuroscience literature, which are shown to exhibit macroscopic processes that are akin to software in human-engineered systems. Overall, this framework enables a deeper understanding of the multi-level structure of complex systems, revealing specific ways in which they can be efficiently simulated, predicted, and controlled.



Causal states

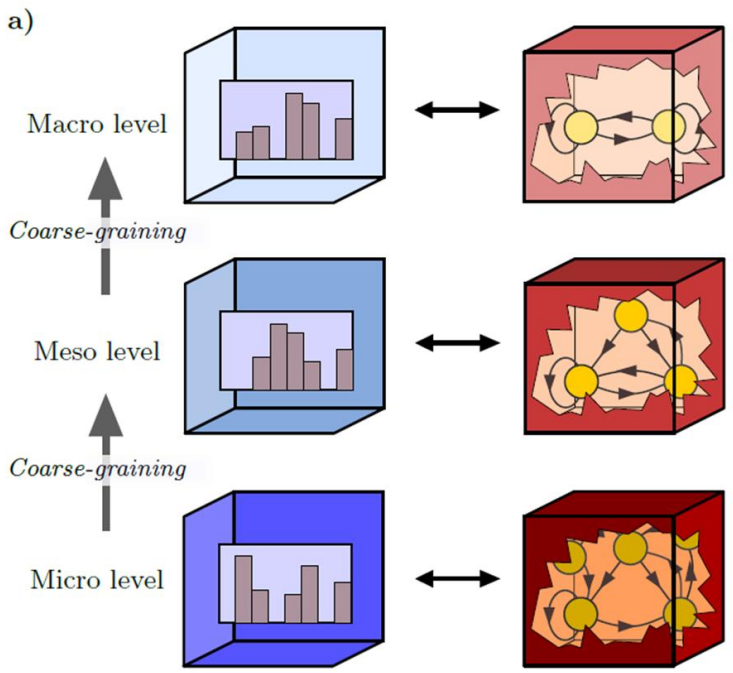
Computational Closure

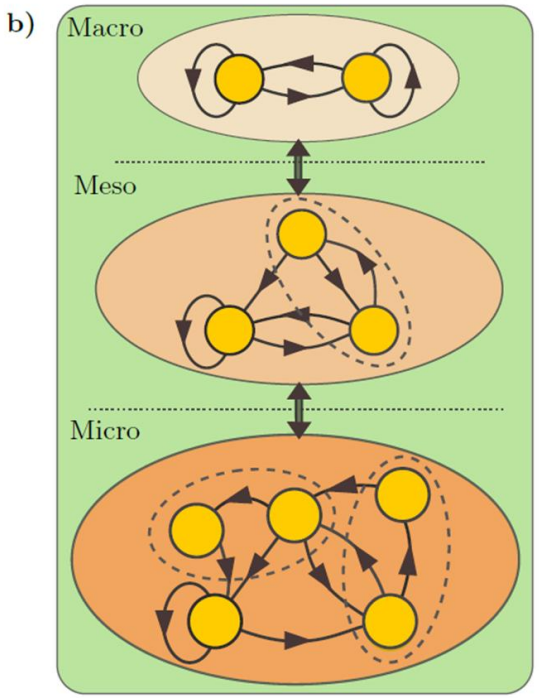
05.07.2024

Rosas et al.: Software in the natural world (2)

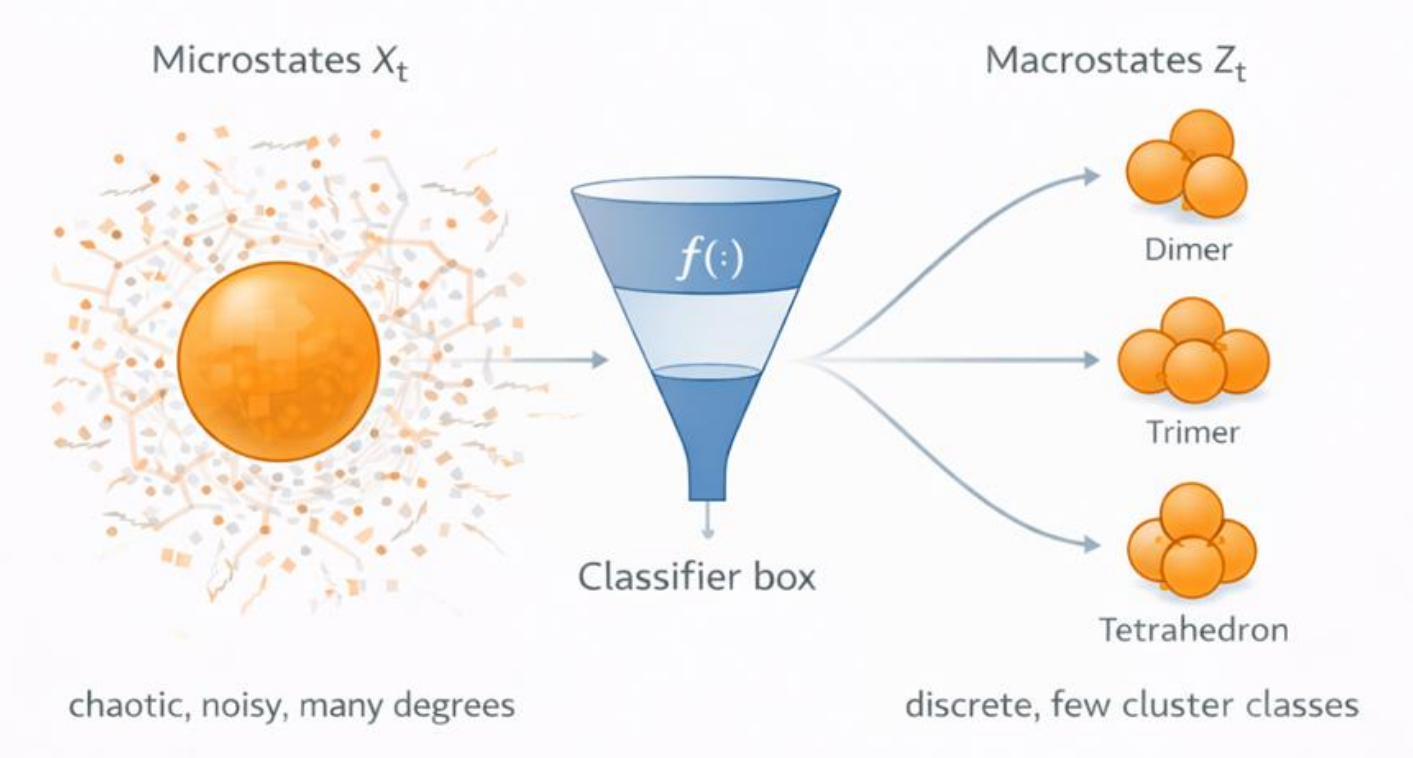
TABLE I. Summary of main concepts

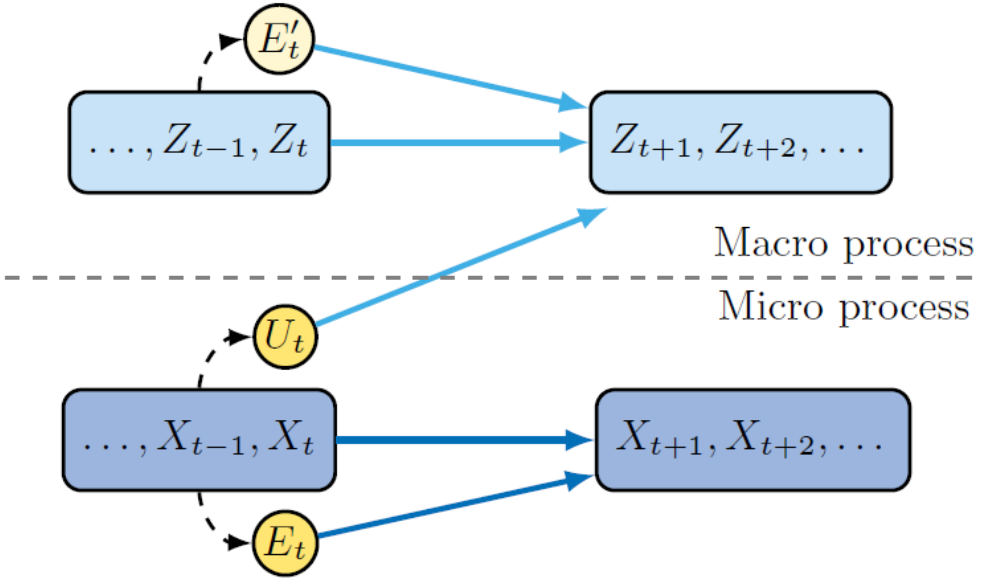
Term	Explanation
Micro level	Basic time-series
Macro level	A coarse-graining of a micro process
ϵ-machine	Model of a process built on the history at the same level
υ-machine	Model of a process built on the history at a level below
Causal states	Hidden states of ϵ- or υ-machines
Information closure	When best predictions of macro can be done from the same macro
Causal closure	When ϵ- and υ-machines coincide
Computational closure	When ϵ-machine of macro is a coarse-graining of ϵ-machine of micro
Lumpability	The ability of a Markov process to remain Markovian after being coarse-grained





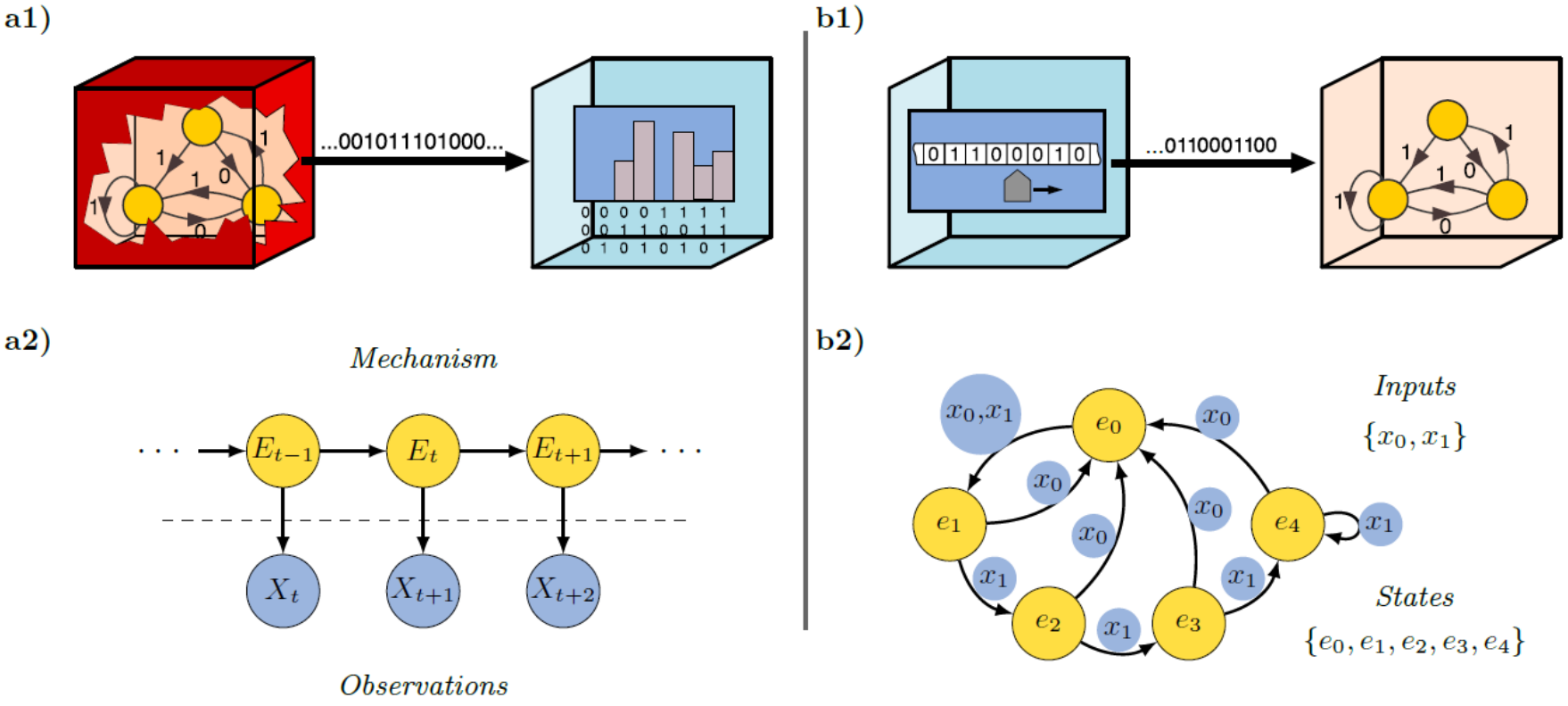
Microstate → Macrostate durch Grobkörnung: X_t zu Z_t





Eine **ε-Maschine** ist das kleinste Modell, das alle für die Zukunft relevanten Informationen über ein dynamisches System speichert. **Minimal vollständige Vorhersagemaschine** eines Prozesses. Das Gedächtnis des dynamischen Systems in komprimierter Form.

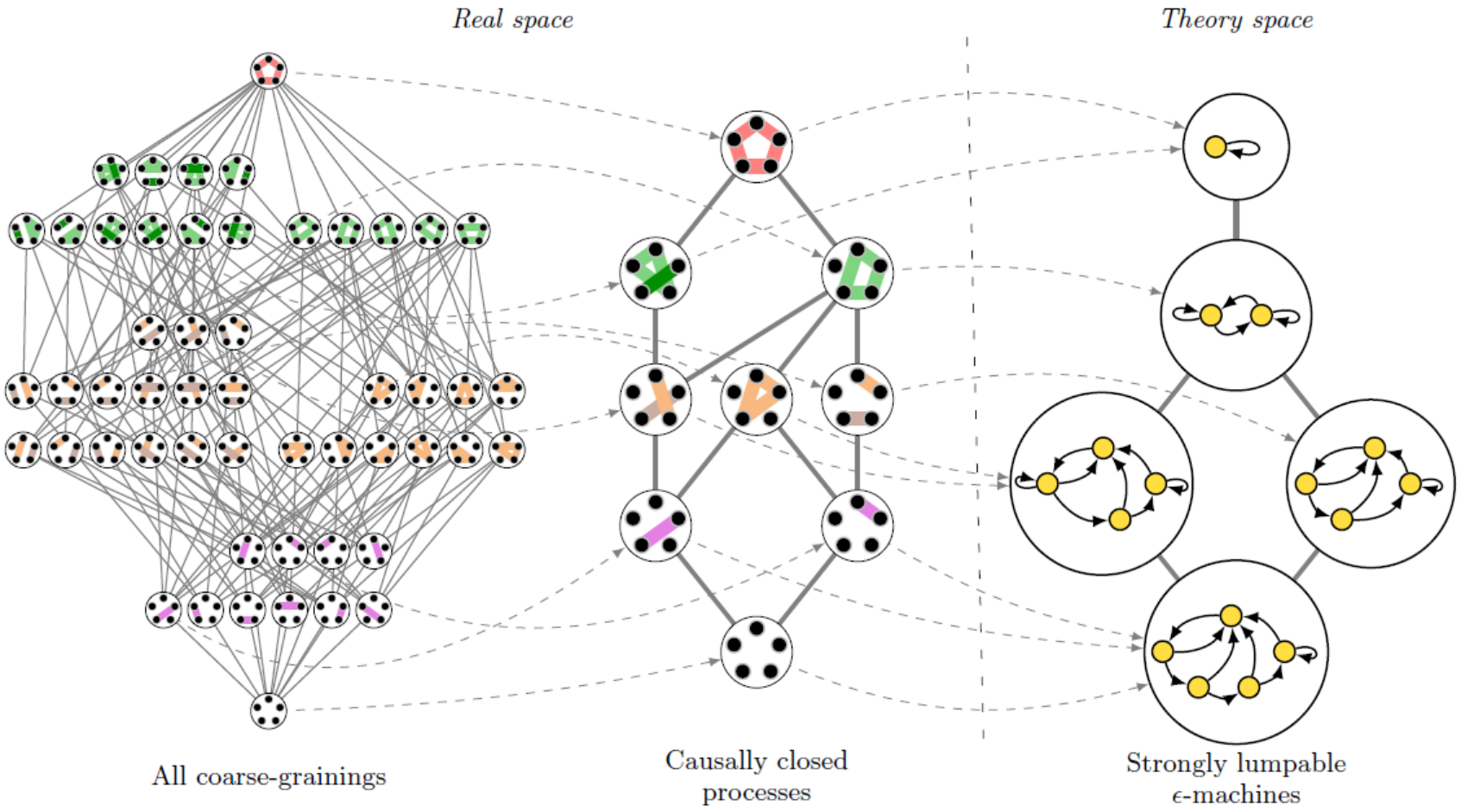
Rosas et al.: Software in the natural world (3)



The two faces of ϵ-machines: Illustration of the dual interpretation of ϵ-machines that establish a bridge between causality and computation. a) **Causal face** and b) **Computational face**.

(**a1**). Technically a hidden Markov process; dynamics that take place on variables Et on a latent state-space,
(**a2**) while generating the observable data Xt.
(**b1**) **ϵ-machines** as discrete automata, where the data corresponds to inputs given by a user driving the system between different states .
(**b2**) Technically a discrete automata with states ek, whose deterministic transitions are governed by the input data xi
Note that (**a1**) focuses on variables (e.g. Xt,Et), while (**b2**) portraits the states that those variables can take (e.g. x0, e0).

Rosas et al.: Software in the natural world (4)



Left : Lattice of all possible coarse-grainings, here illustrated for the case of a process that can take five possible values.
Center: Sub-lattice of only those coarse-grainings that are causally/informationally closed.
Right: Lattice of strongly-lumpable coarse-grainings of the ϵ-machine of the microscopic level.
Only the last lattice provides a minimal blueprint that highlights the distinct computational processes, and distinguishes which computations take place at what level.

The multiple hiearchies describing multi-level computations in a complex system.

05.07.2024

Informational Closure:
„Reicht der Makrozustand zur Vorhersage?“

Kernfrage: Sagt der **Makrozustand Z_t** die **nächste Makro-Zukunft** fast genauso gut voraus wie die vollständigen **Mikrodetails X_t**?

Ergebnis: In den Casimir-Lifshitz-dominierten Abstandsbereichen L (= 2-40 nm, $U_{CL} \approx$ 4-25 $k_B T$) enthält der Makrozustand fast die gesamte relevante Information für seine eigene Zukunft. Die Mikrodetails liefern out-of-sample keinen Zusatznutzen.

Einschränkung: Das gilt nicht bei hoher Noise! Sondern nur bei kleinen und mittleren thermischen Fluktuationen! Eine freie Protozelle benötigt etwa 2.3 Sekunden, um durch Brownsche Bewegung einen eigenen Durchmesser (1.000 nm) zurückzulegen.



Computational Closure:
„Gibt es eine kompakte Makro-ε-Maschine?“

Kernfrage: Kann die **Dynamik der Protozellcluster** durch ein kleines Makromodell oder eine **ε-Maschine** beschrieben werden, ohne die Mikrodaten mitzuschleppen?

Ergebnis: Die Casimir-Lifshitz-stabilisierten Protozellen-Cluster besitzen eine eigene kompakte Dynamikbeschreibung. Das ist der stärkste Hinweis auf eine **proto-softwareartige Makroebene**.

Einschränkung: Das Computational Closure ist überwiegend **kurzfristig** (**Sekundenbereich**); ein ausgeprägtes Langzeitgedächtnis wird nicht beobachtet. Wir sehen aber eine **kompakte, eigenständige Makro-Dynamik**. Tetraeder => Lebensdauer exponentiell $e^{\Delta E/kBT}$



Casual Closure:
„Wirkt der Makrozustand kausal stabilisierend?“

Kernfrage: Wenn ich **verschiedene Mikrokonfigurationen** nehme, die alle denselben Makrozustand bilden, führen sie **nach einer Intervention zu ähnlichen Makro-Zukünften?**

Ergebnis: Nur teilweise. Bei schwacher Störung zeigen die Casimir-Lifshitz-dominierten Abstandsbereiche eine gewisse makro-kausale Trennung. Bei stärkerer Störung werden die Unterschiede innerhalb desselben Makrozustands größer.
Das heißt: Der Makrozustand schirmt die Mikrodetails noch nicht vollständig ab.

Einschränkung: Die Simulation zeigt **keine strenge kausale Closure**, sondern eine **heuristische, interventionelle Evidenz für partielle makro-kausale Stabilität**.

Ausblick: Tetraeder sollten aufgrund ihrer sechs Kontaktkanten eine deutlich stärkere Causal Closure zeigen als einfache Dimere.

Ergebnisse und Take-Home-Message:

Casimir–Lifshitz-stabilisierte Protozellcluster zeigen unter niedriger bis moderater Noise eine **emergente Makroebene**, die informationsmäßig und computational weitgehend geschlossen ist: Sie ist vorhersagekräftig und kompakt modellierbar.

Casimir–Lifshitz-Kräfte erzeugen nicht nur physikalische Protozellen-Cluster,
sondern eine **emergente Makroebene** mit:

- **Syntax:** stabile Zustandsstrukturen,
- **Semantik:** vorhersagbarer Informationsgehalt,
- **Pragmatik:** kausaler Wirksamkeit auf zukünftige Dynamik.

Von physikalischen Fluktuationen über Casimir–Lifshitz-stabilisierte Protozellen-Cluster entsteht eine **proto-informationelle Ebene**, auf der **Syntax**, **Semantik** und erste **pragmatische Wirksamkeit** bereits vor RNA, Genetik und biologischer Evolution beobachtbar werden können.


1 Thermisches Rauschen
2 Casimir-Lifshitz-Kopplung
3 stabile Cluster
4 stabile Attraktoren
5 ε-Maschinen
6 informational closure
7 proto-semantische Zustände
8 proto-pragmatische Aufgaben
9 erste präbiotische Informationsverarbeitung
A
B
C
Zufällige thermische Fluktuationen auf mikroskopischer Ebene
$\langle x^2 \rangle = 2\,D\,t$
Vakuumfluktuationen erzeugen anziehende Kräfte zwischen Körpern
$F = -\frac{\pi^2 \hbar c A}{240 d^4}$
Selbstorganisation durch nicht-kovalente Wechselwirkungen
$\Delta G < 0$
Dynamische Systeme entwickeln stabile Zustände (Attraktoren)
$\frac{dx}{dt} = f(x)$
Minimal ausreichende prädiktive Modelle der eigenen Dynamik
$P(\text{Zukünftig}) = P(\text{Zukünftig} \mid \text{Vergangenheit})$
Informationsaustausch hauptsächlich innerhalb des Systems
$I(X_{\text{innen}}; X_{\text{außen}}) \approx 0$
Zustände repräsentieren Unterscheidungen mit informativem Gehalt
$I(\text{Zustand}; \text{Umwelt}) > 0$
Zielgerichtete Selektion und Ausführung von Handlungssequenzen
$P(\text{Erfolg}) > P(\text{Zufall})$
Integration von Wahrnehmung, Gedächtnis und Handlung in präbiotischen Systemen
Informationsverarbeitung vor dem Leben

Thank you very much for your attention.

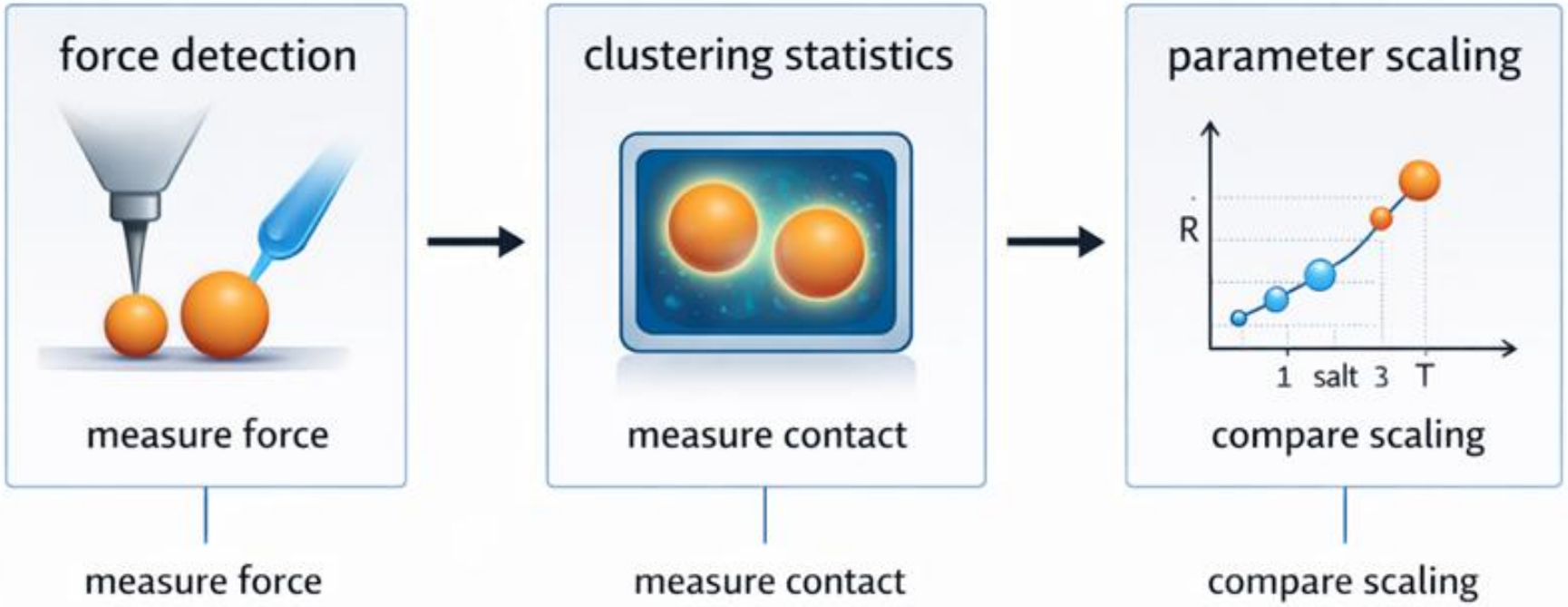
Are there any questions?



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Experimental pipeline linking force measurements to observable clustering and quantitative scaling:



Attractive Casimir–Lifshitz Forces as a Universal Driver of Prebiotic Protocell Aggregation and Cluster Formation

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Abstract—**How fragile, RNA-free protocells could have formed stable clusters in a hot, saline and dynamically fluctuating prebiotic environment remains unresolved. Classical explanations—such as hydrophobic effects, Derjaguin–Landau–Verwey–Overbeek (DLVO) interactions and thermal noise—fail in the relevant 5–200 nm separation range. Here we propose that attractive Casimir–Lifshitz forces act as a universal and physically unavoidable coupling mechanism between protocells. Starting from the exact sphere–sphere solution in the classical (high-T) limit and a Derjaguin (PFA) approximation, we show that Casimir–Lifshitz attraction decays algebraically as $1/L^2$ under prebiotically plausible conditions (protocell radii 100–1,000 nm; separations 5–200 nm; 50–200 mMol salt; 20–90 °C) and can exceed exponentially suppressed DLVO contributions beyond a few nanometres. From this framework we derive testable predictions—including stronger clustering of PMBCs relative to fatty-acid vesicles, enhanced effective adhesivity of larger protocells and contact lifetimes on the order of minutes—and outline a three-stage experimental roadmap to evaluate Casimir–Lifshitz forces as a realistic driver of prebiotic protocell cluster formation.**

theory [3] are strongly screened under realistic ionic conditions and rapidly overwhelmed by thermal noise at the k_BT scale.

This gap motivates a central question: which physical mechanism could have supported reproducible, mesoscale, and non-chemical encoded protocell cooperation prior to genetic or metabolic specialization? Here we propose that attractive Casimir–Lifshitz (CL) interactions [1][2] —arising from quantum and thermal electromagnetic field fluctuations—constitute a physically unavoidable, material-dependent aggregation mechanism operating precisely in the regime where classical colloidal forces fail. Unlike DLVO-type interactions, CL forces do not rely on surface charge or molecular specificity and persist across nanometre-to-submicrometre separations in electrolyte environments.

In this work, we develop a quantitative theoretical framework for CL-mediated protocell interactions, establish their relevance relative to classical screened forces, and translate the resulting scaling laws into experimentally testable



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Emergent Information Formation in Prebiotic Protocell Clusters: A Computational Mechanics Framework of ε-Machines and Attractor Memory

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Abstract- **Casimir–Lifshitz forces generate an unavoidable, long-range attraction between protocells under prebiotically realistic conditions. This interaction stabilizes mesoscale clusters such as tetrahedra, octahedra, and 13-cell icosahedra. These highly symmetric assemblies act as persistent macrostates whose transitions remain reproducible despite microscopic noise. A physics-guided coarse-graining yields a well-defined mesodynamics that can be represented as an ε-machine: a small deterministic automaton whose causal states correspond to cluster attractors and whose transitions encode ordered reconfiguration pathways. The theory of Rosas et al. ("Software in the natural world") shows that such systems can become informationally, causally, and computationally closed, thereby forming an autonomous proto-software layer. In this framework, prebiotic information does not arise from polymers but from attractor-based memory and structured transition dynamics in a purely physical cluster process.**

In Figure 1, a schematic representation of two spherical protocells with radii R_1 and R_2 is shown, separated by a minimal surface-to-surface distance L in saline aqueous solution (primordial soup). This sphere–sphere geometry defines the fundamental configuration used to model Casimir–Lifshitz interactions between protocell membranes under prebiotic conditions.

Casimir–Lifshitz forces arise from quantum and thermal fluctuations of the electromagnetic field between material interfaces, as first described by Hendrik Casimir [3] and generalized by Evgeny Lifshitz [4]. Unlike chemical bonds or electrostatic interactions, they do not rely on charges, receptors, or molecular specificity, but on the dielectric properties of the interacting materials and medium. In saline, thermally active environments, these forces act over nanometre-to-submicrometre distances and decay



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From Physical Difference to Meaning: A Constructor-Theoretic Framework for Prebiotic Information in Casimir-Lifshitz-Coupled Protocell Clusters

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Abstract- **This paper develops a physical framework for the prebiotic emergence of information and meaning. Building on Constructor Theory, we define information as a reproducible physical difference and meaning as a difference with stable functional consequences. Casimir–Lifshitz–coupled protocell clusters serve as a minimal model that exhibits reproducible attractors, ordered transitions, and autonomous task structures. We show that such clusters carry both informational states (e.g., distances, geometries, gradients) and meaningful states that regulate prebiotic tasks such as approach, exchange, or stabilization. This approach integrates physical mechanisms, computational mechanics, and early proto-semantic functions into a coherent account of information formation before biology.**

The structure of the paper is as follows: In Section II, we introduce the motivation, conceptual gap, and guiding research questions, framing prebiotic information and meaning within a constructor-theoretic perspective. Section III reviews related work on semantic information, protocells, and Constructor Theory, positioning our approach relative to existing biological and computational models.

Section IV outlines the core principles of Constructor Theory, with a focus on tasks, constructors, and the physical definition of information. In Section V, we formalize information as a reproducible physical difference and demonstrate how protocell cluster attractors satisfy constructor-theoretic information criteria.



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Physical Origin of Proto-Information: Syntax, Semantics and Pragmatic Emergence in Prebiotic Protocell Clusters

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Abstract- **The emergence of information and meaning before genes or enzymes existed is a key open problem in prebiotic research. This paper develops a physics-based framework for the rise of early information in protocell systems. We show that Casimir–Lifshitz interactions, acting under plausible primordial-soup conditions, generate stable dimers and ordered clusters in which nanometer-scale resonant gaps create structured fluctuation spectra and proton-rich reaction zones. On this basis, Computational Mechanics is used to describe the syntactic layer via ε-machines, functional semantics via lumping operations, and the pragmatic layer via emergent macro-ε-machines that execute cluster-level tasks in the sense of Construction Theory. Together these elements yield a coherent model of proto-biological meaning: one that explains prebiotic information structures without genes, enzymes, or biochemical networks and identifies a physical precursor to functional organization.**

syntactic, semantic, and pragmatic dimensions into a coherent framework of proto-biological meaning.

In addition, we integrate the 'Software in the Natural World' framework of Rosas et al. [4] with the Constructor Theory of Information [5] developed by Deutsch and Marletto, applying both to the hierarchical and task-based organization of protocell cluster dynamics.

The structure of the paper is as follows: In Section II, we review existing approaches to semantic information, Computational Mechanics, and Constructor Theory, highlighting the absence of a unified, physically grounded framework for prebiotic meaning. Section III introduces the physical and conceptual background of prebiotic protocell systems and formulates the central research questions concerning the emergence of syntax, semantics, and pragmatics prior to genetics. Section IV develops the physical



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Quantum Field-induced Proton H^+ Gradients in Prebiotic Protocell Clusters

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Abstract- **The origin of the first hydrated proton (H^+) gradients before the rise of biological proton pumps or membrane-bound enzyme complexes remains a central unresolved problem in prebiotic research. Here we show that protocell clusters, formed under saline primordial-soup conditions, are not only stabilized by Casimir–Lifshitz attraction but also generate selective modulation of dynamic Matsubara modes within their nanometer-scale gaps. Because the static mode (n = 0) is fully suppressed by Debye screening, the remaining dynamic modes create geometry-defined resonance zones with lowered vacuum energy. These zones exponentially enrich hydrated protons (H^+) through Casimir–Polder coupling. Dimers act as one-dimensional proton (H^+) channels, while tetrahedral clusters produce three-dimensional proton funnels. The resulting quantum-field-induced proton reservoirs offer a physically plausible precursor to chemiosmotic (H^+) gradients and may have provided the first energetic coupling mechanism between early protocells.**

Massoth [3] shows that protocell clusters can generate reproducible differences and functional meaning states. These proto-functional states form the conceptual basis for this paper, where proton gradients appear as early functional signals emerging from physical constraints.

Massoth [4] reveals that Casimir-Lifshitz-shaped resonance zones and ε-machines create syntactic, semantic, and pragmatic layers of early information. This multi-layered framework supports this paper, which interprets proton reservoirs as energetically encoded proto-semantic structures within nanoscale reaction zones.

Structure of the paper: Section II introduces the physical framework of Casimir–Lifshitz–driven protocell clustering and defines the relevant geometric configurations. Section III develops the quantum-field mechanism underlying proton accumulation in nanoscale resonance zones and provides order-of-magnitude estimates. Section IV discusses the



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Prebiotic Sustainability in the Nanoworld: A Physical Framework for Early Protocell Cluster Stability

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Abstract- **This paper develops a physically grounded concept of prebiotic sustainability in the nanoworld. It shows how early protocell clusters were able to form stable, persistent, and regenerative states of order through fluctuation-induced forces, nanofluidic flows, and proton-based gradients—before genes or enzymes. The model describes sustainability as an emergent property of open non-equilibrium systems and integrates three dimensions: structural, energetic, and informational sustainability. The resulting framework enables testable hypotheses about stability, gradient duration, and regeneration processes. Thus, the approach opens up new experimental avenues for protocellular self-preservation and provides criteria for evaluating early prebiotic scenarios.**

emergent stabilization of such multilayered information processes.

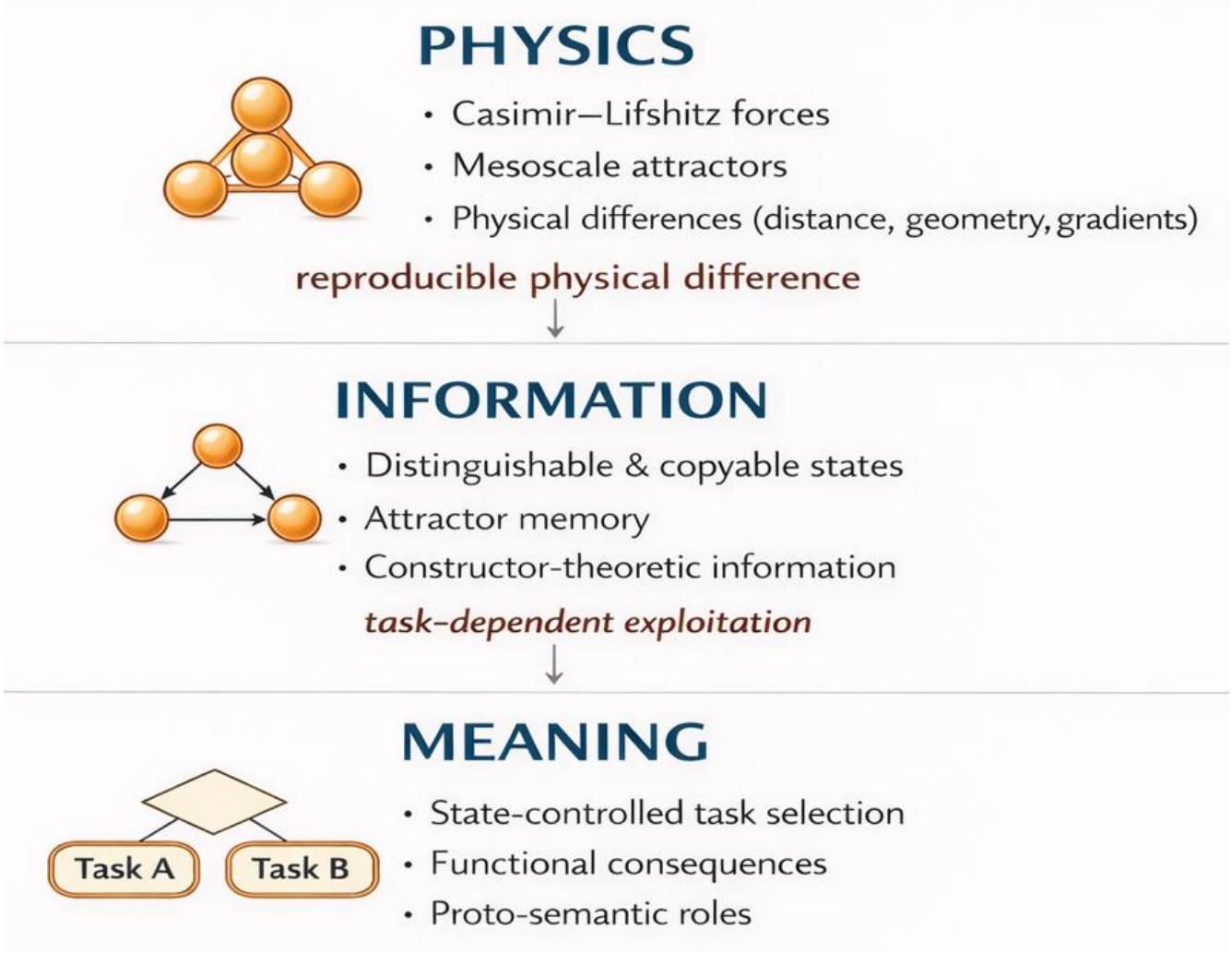
Massoth [5] identifies Matsubara-mode selection and Casimir–Polder coupling as sources of stable proton (H^+) gradients in dimers and tetrahedra. These quantum-field-driven energy structures form the energetic core that this paper develops into "energetic sustainability" and links to experimental tests.

The structure of the paper is as follows: Section I presents the motivation, state of the art, and introduces prebiotic sustainability as a physical non-equilibrium principle. Section II develops the nanophysical baseline model and derives



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The Grand Synthesis: From Physics to Meaning



Meaning emerges when
reproducible physical differences systematically control functional tasks.

Unified Architecture (1)

Pragmatics (=Tasks or actions enabled by those states)

Macro-ε-machines implement repeatable tasks

Semantics (=Functional meaning of those states)

Functional lumping of syntactic states

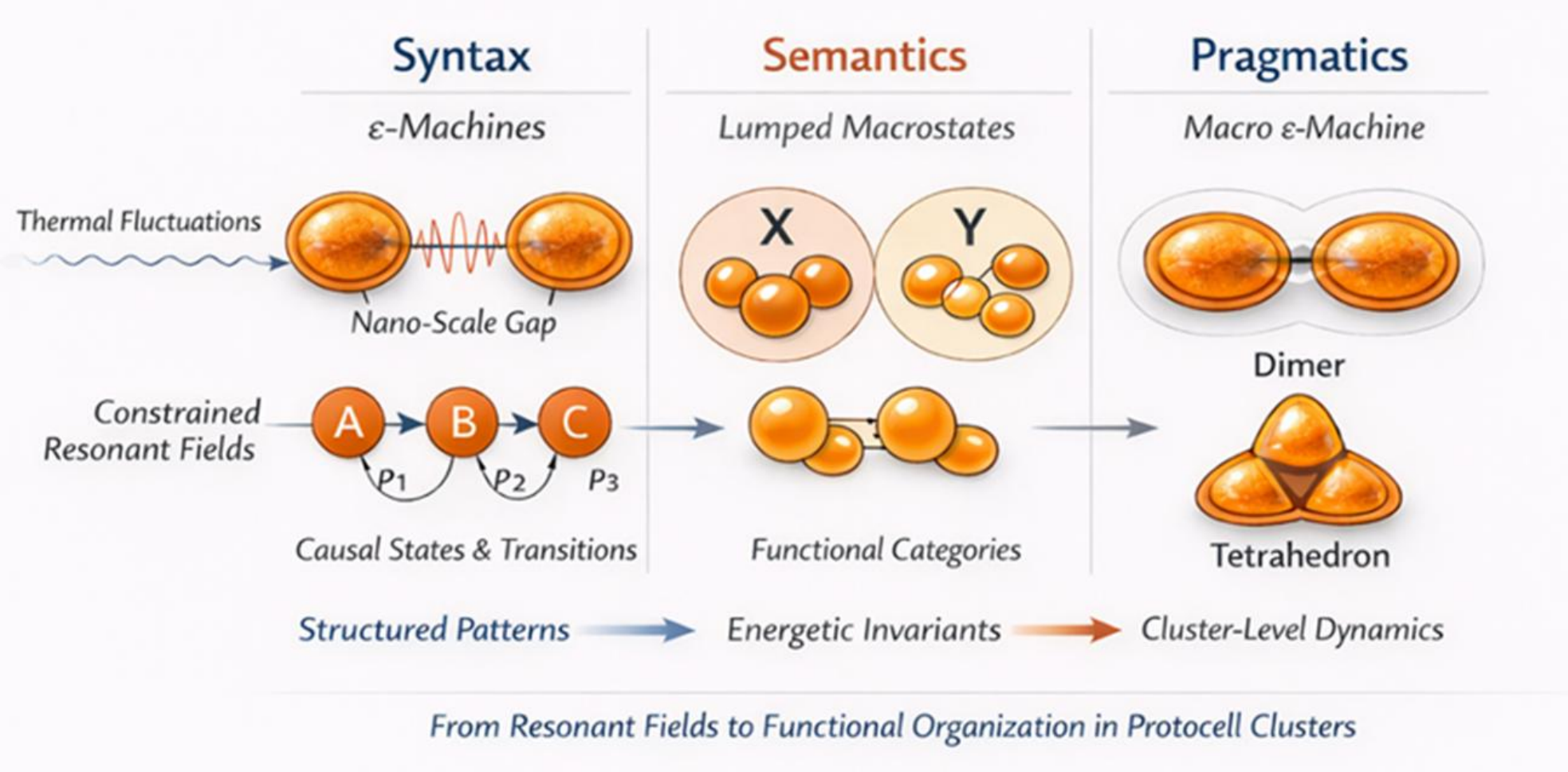
Syntax (=Structure of patterns or states)

ε-Machines causal states

Physics

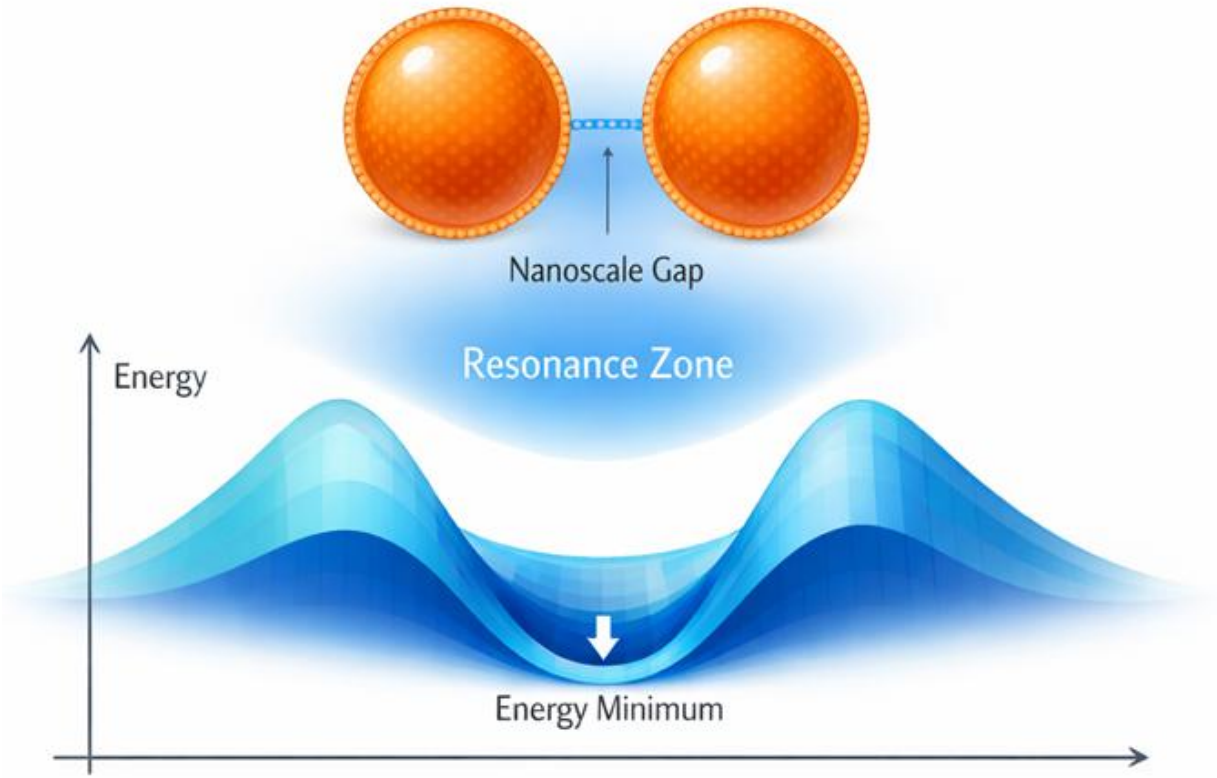
Casimir-Lifshitz coupling structures fluctuation spectra

Unified Architecture (2)



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Fluctuation Energy Resonance Zone:
Proto-Bio-Energetik

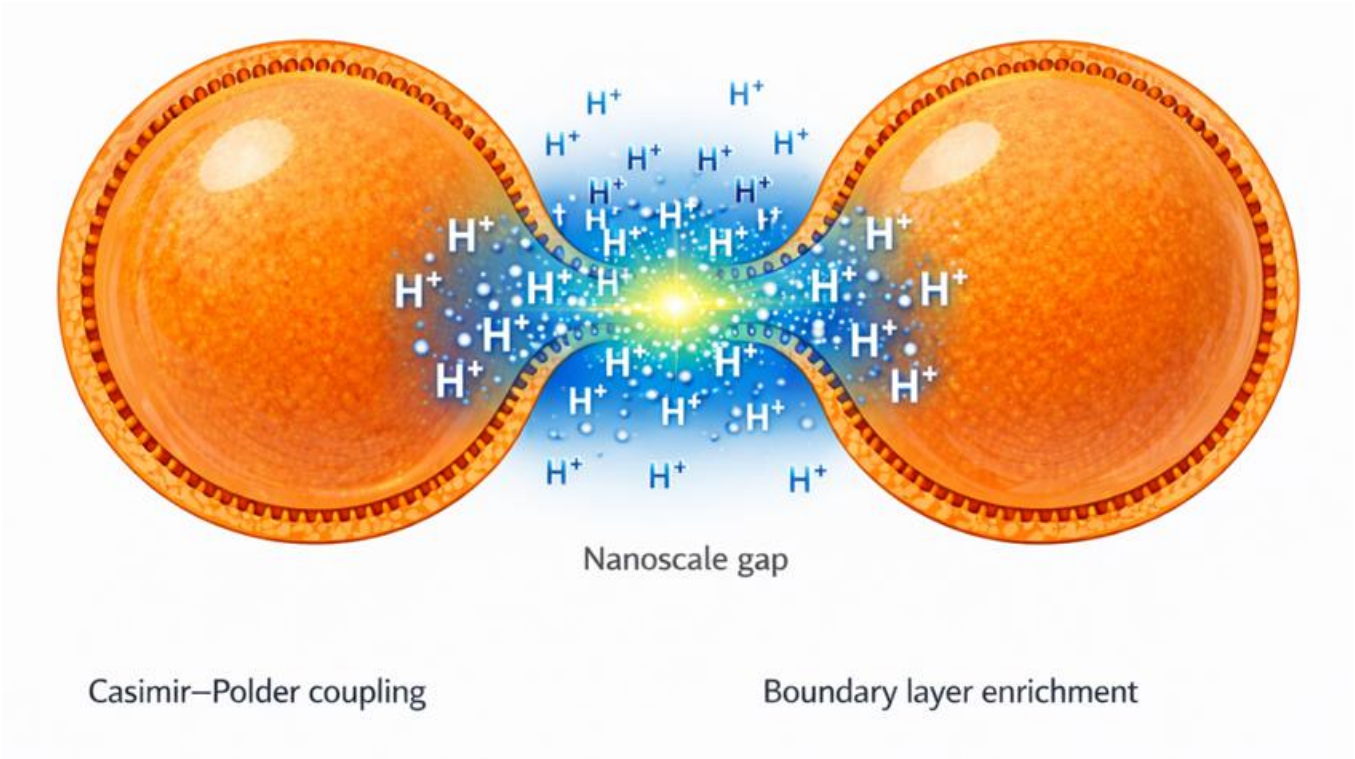


Why hydrated protons accumulate:

- **Hydrated protons** are highly polarizable clusters in water.
- They couple to the modified electromagnetic mode density in the gap.
- **Casimir–Polder coupling** creates an **effective potential U_CP(z)** with **minima in resonance zones**.

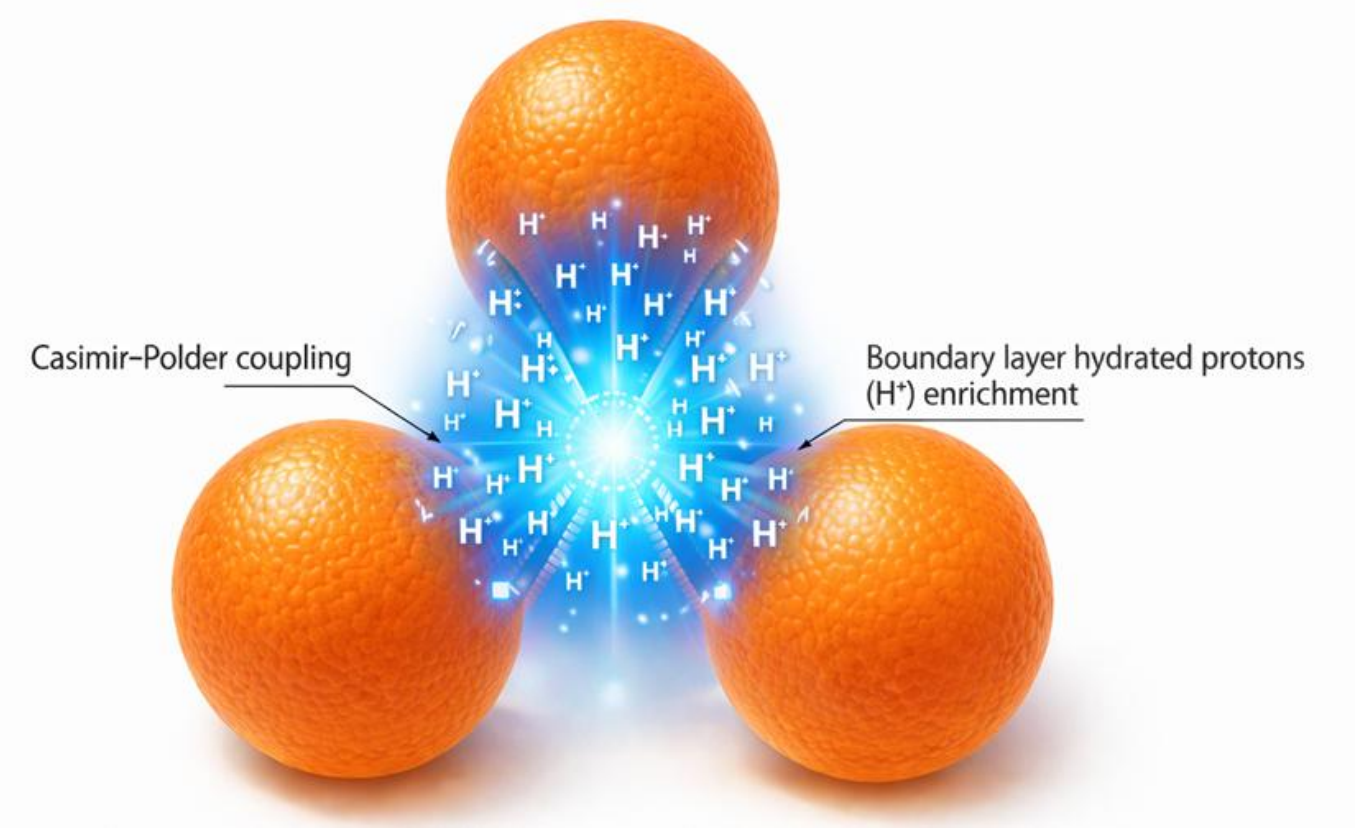
Für ein Dimer wäre der Gap ein bidirektionaler protonischer Speicherkanal: $C_1 \leftrightarrow G_{12} \leftrightarrow C_2$

Proton Accumulation Mechanism:



Im Gap bildet sich eine fluktuationsarme korrelierte Nanodomäne

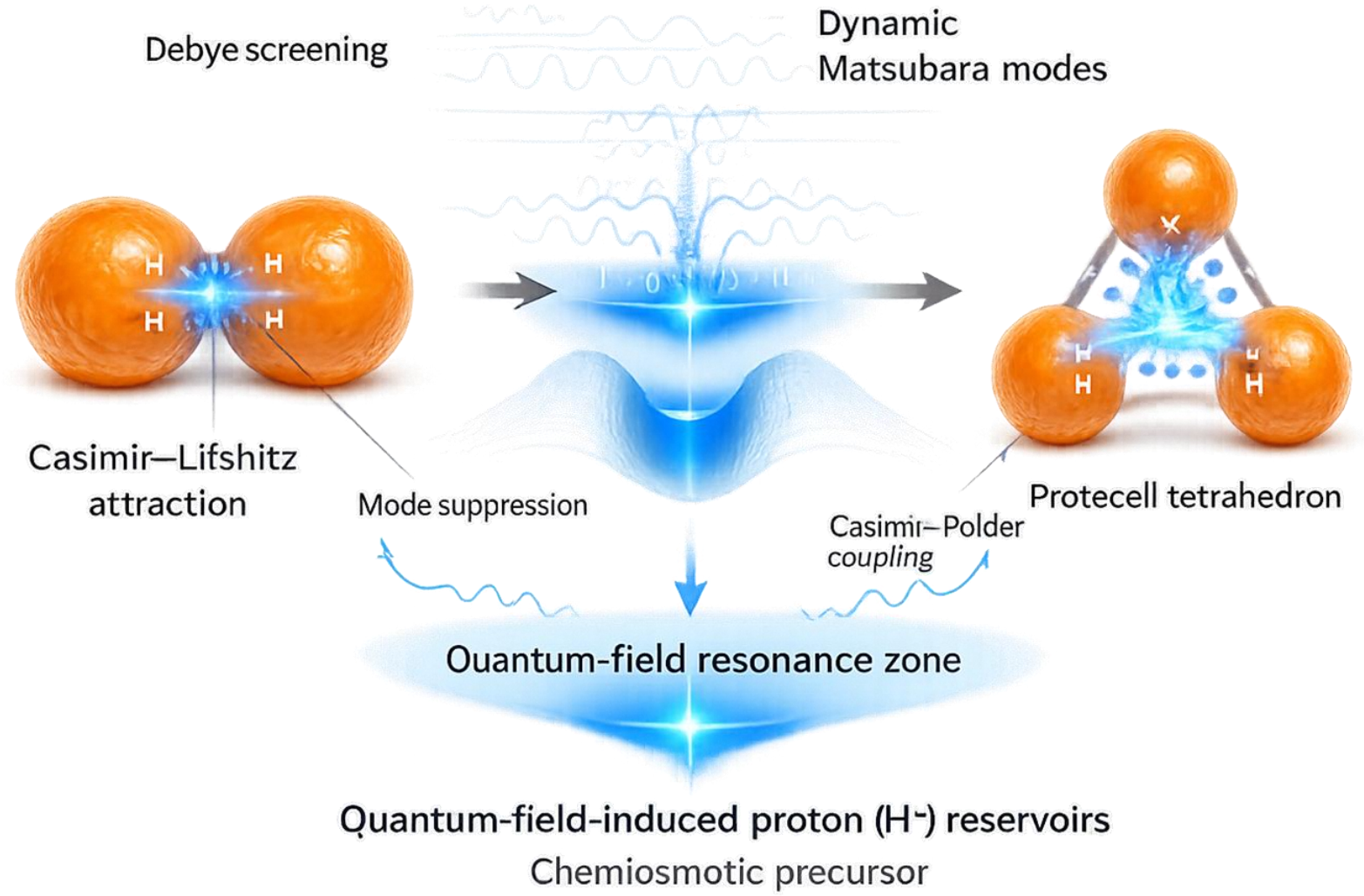
Tetraeder-Protozell-Cluster: 3D-Protonen-Knotenpunkt (Trichter)



- Four interfaces confine modes more strongly than a dimer.
- Multiple scattering produces a deep central resonance well.
- **Result:** a **3D proton hub (funnel)** — a proto-chemiosmotic reservoir, mainly on the surfaces of the membranes.

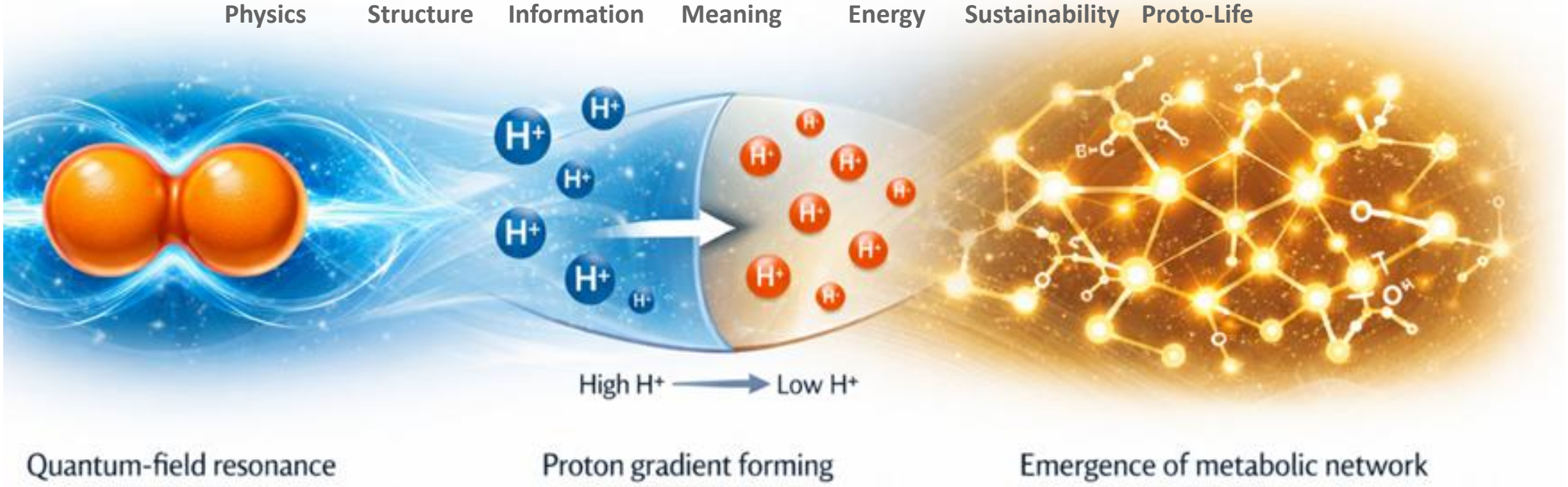
Im Tetraeder entstehen **sechs gekoppelte Nanogaps:** $G_{12}, G_{13}, G_{14}, G_{23}, G_{24}, G_{34}$

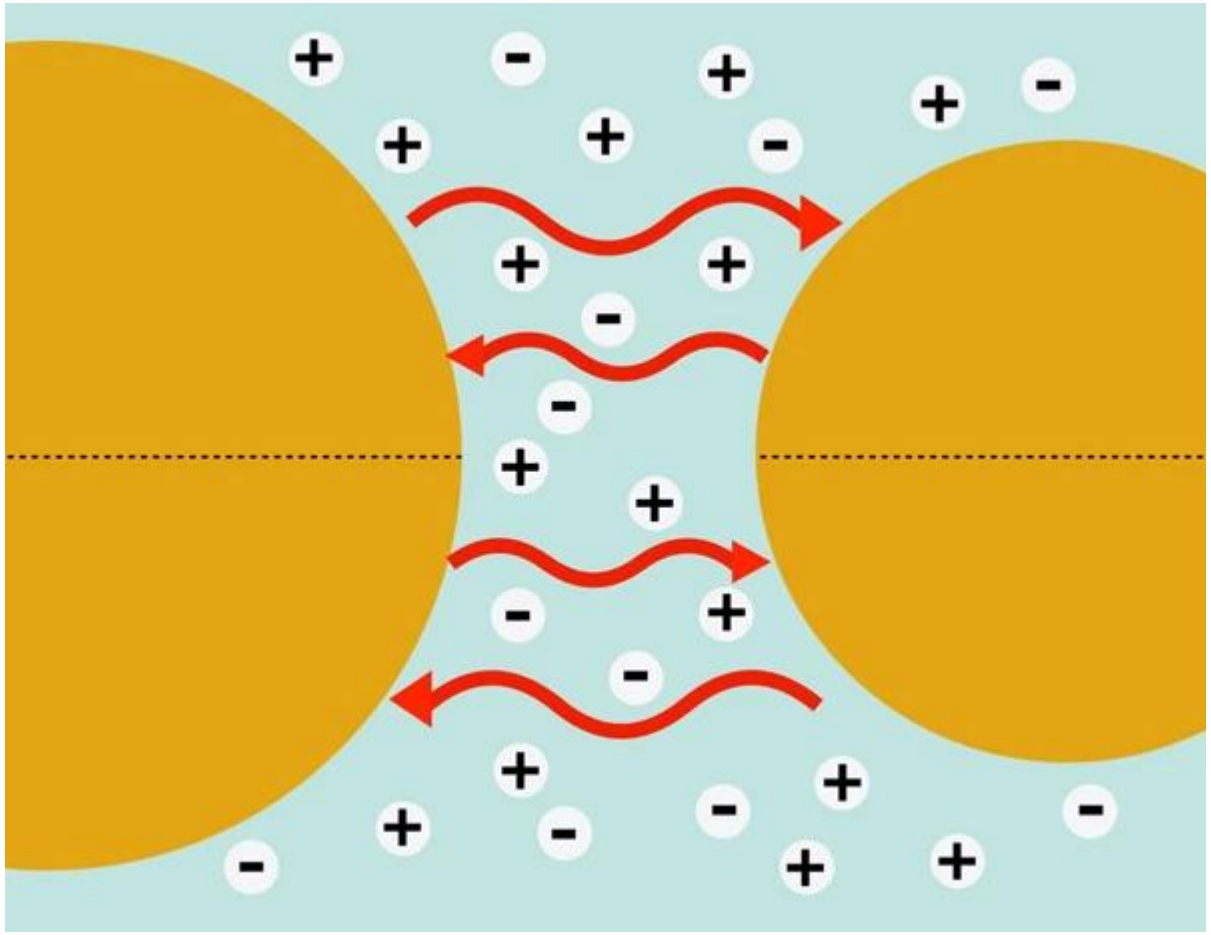
Protongradienten sind Energiequellen für Protozellen

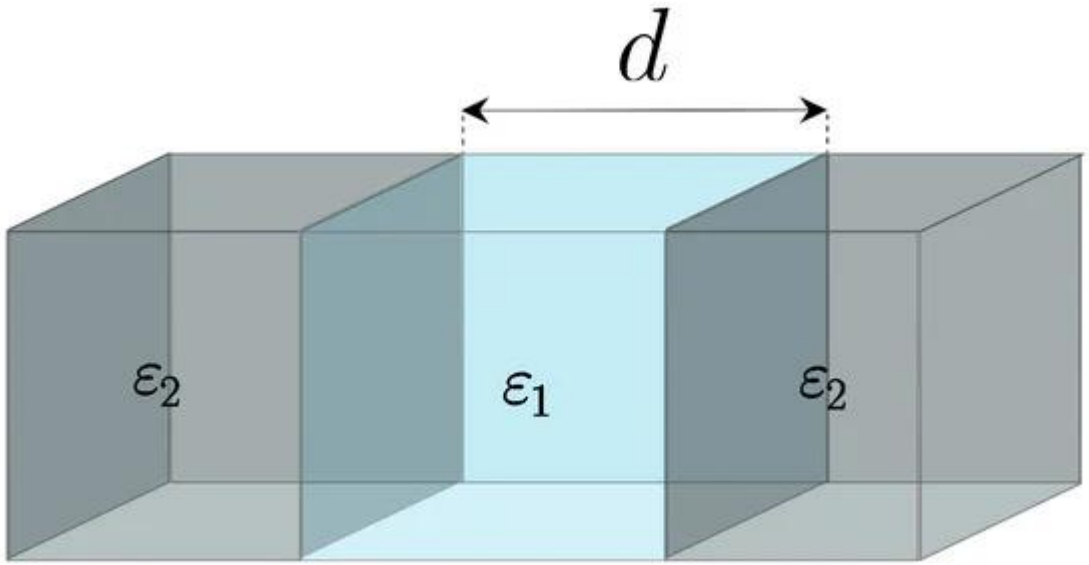


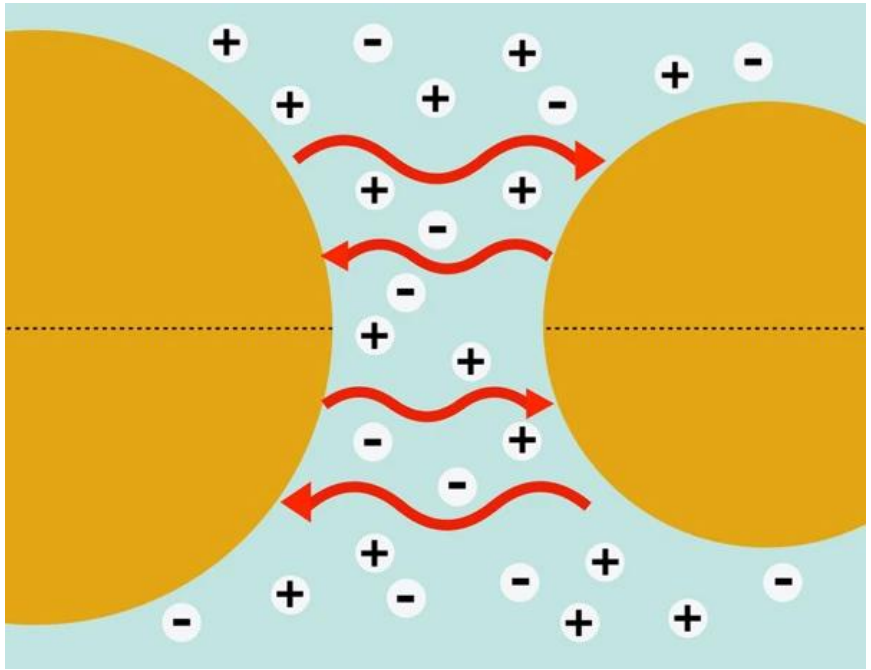
Bridge: From Physics and Proton Gradients to Emergence & Sustainability:

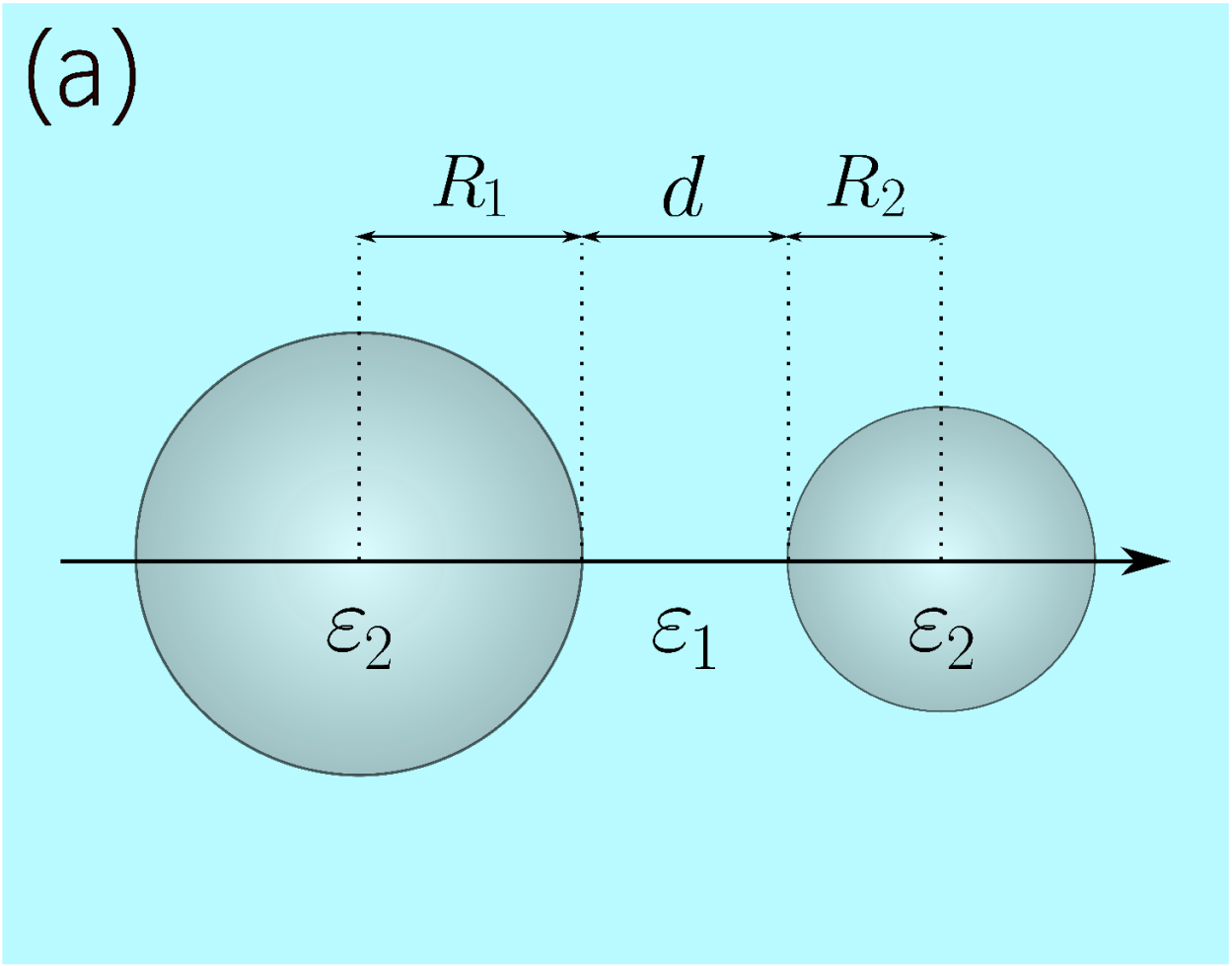
1 Physics | 2 Structure | 3 Information | 4 Meaning | **5 Energy** | 6 Sustainability | 7 Proto-Life

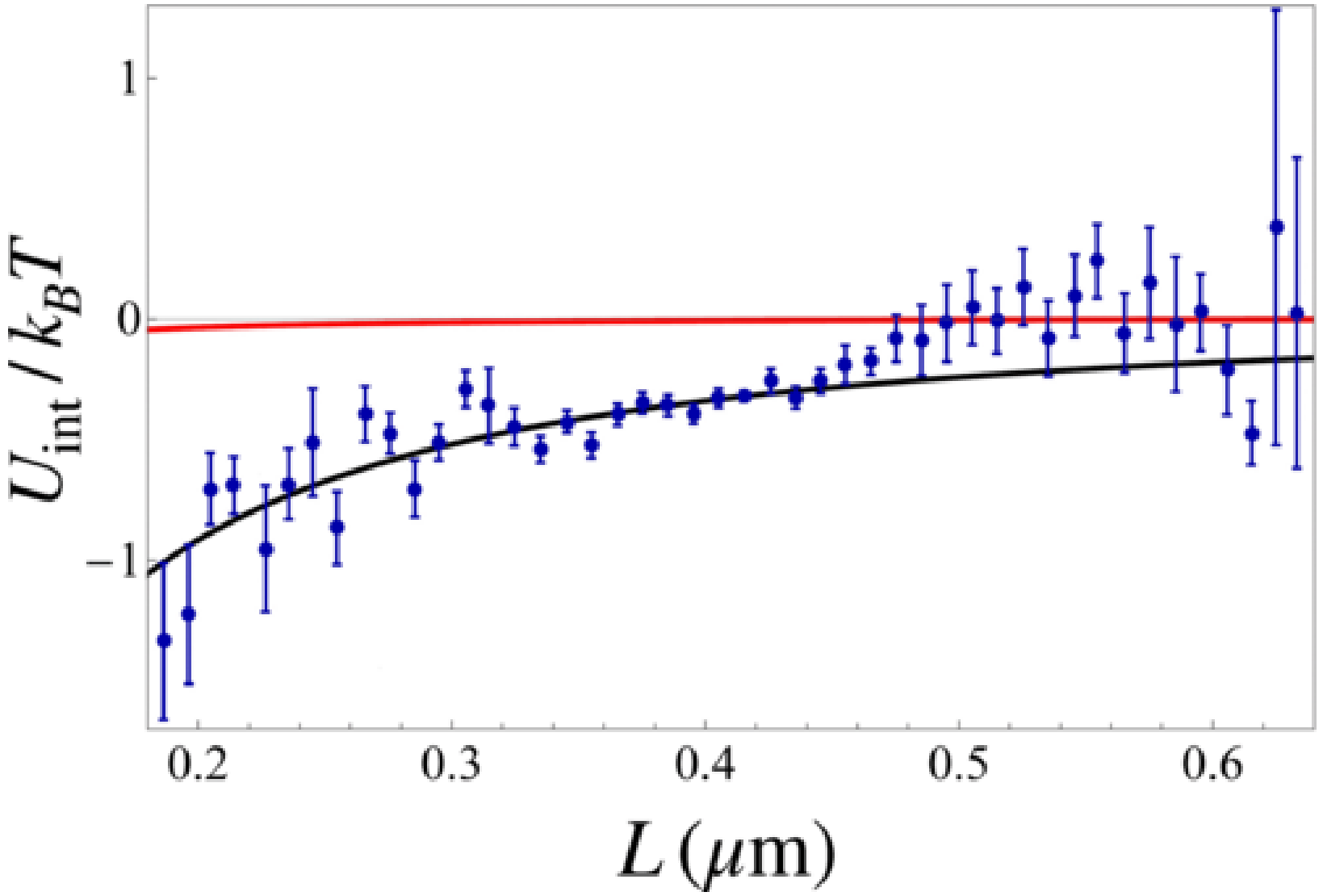





d
ε2
ε1
ε2




(a)
R_1
d
R_2
ε_2
ε_1
ε_2



Casimir energy in units of kBT versus distance for the sample with a high salt concentration: experiment (points with error bars representing the standard error of the mean) and theory either **with (black)** or without (red) the unscreened contribution from TM modes in the limit of zero frequency. The salt concentration of n_∞=0.22M, corresponding to a screening length λ_D=0.64 nm, is such that the double-layer interaction is completely suppressed in the distance range probed experimentally.

Thank you very much for your attention.

Are there any questions?



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